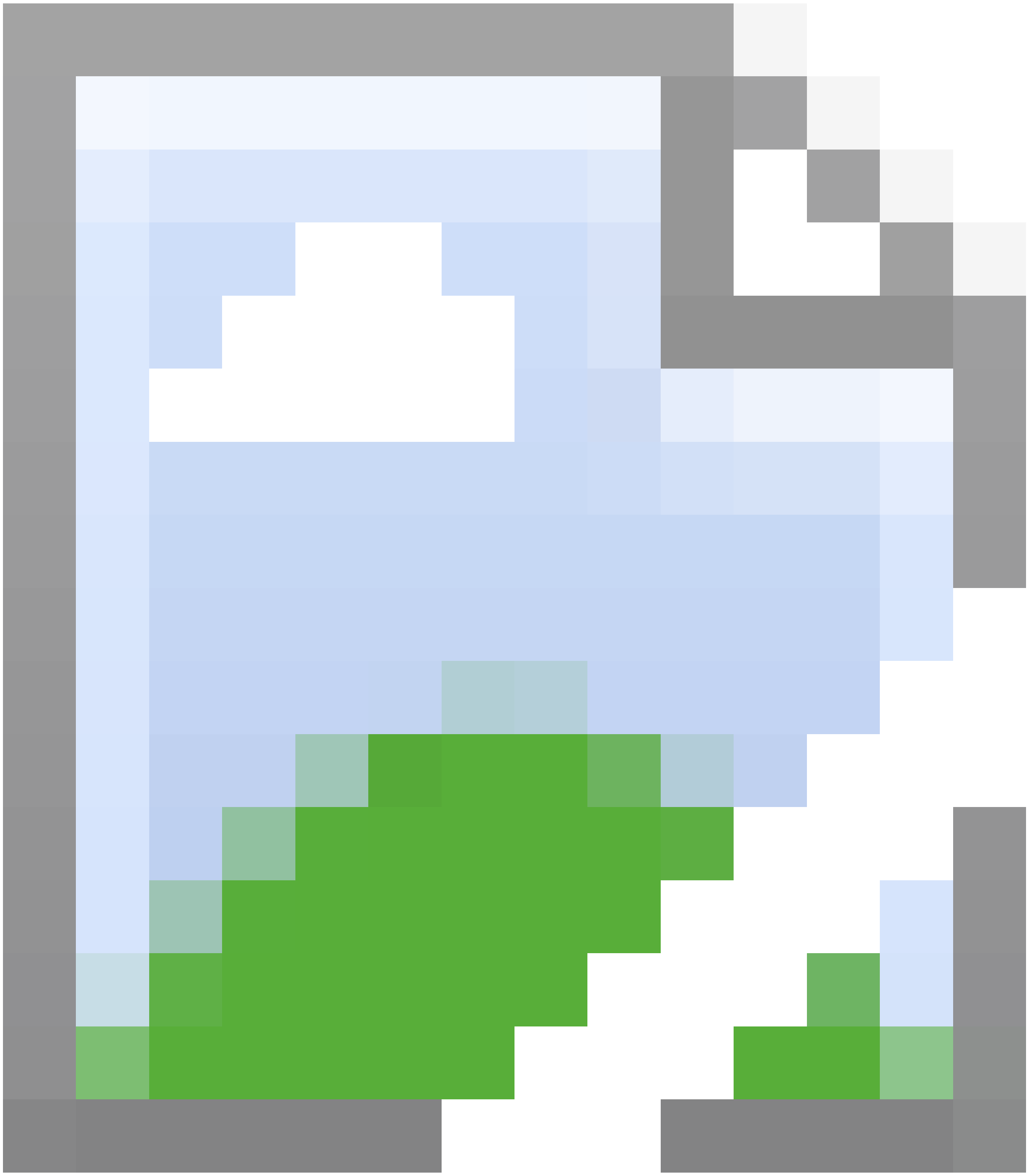
Inferential and Predictive Statistics for Business

Module 3

ILLINOIS
Gies College of Business



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Preface

Thank you for choosing a Gies eBook.

This Gies eBook is based on an extended video lecture transcript made from Module 3 of Professor Fataneh Taghaboni-Dutta's *Inferential and Predictive Statistics for Business* on Coursera. The Gies eBook provides a reading experience that covers all of the information in the MOOC videos in a fully accessible format. The Gies eBook can be used with any standards-based e-reading software supporting the ePUB 3.0 format.

Each Gies eBook is broken down by lessons that are navigable using our e-reader's table of contents feature. Within each lesson the following sequence of content will always occur:

- Lesson title
- A link to the web-based videos for each lesson (You must be online to view.)

Within the lesson, every time there is a slide change or a switch to the next informative video scene, you will be presented with:

- Thumbnail image of the current slide or video scene
- Any text present on the slide in the video is recreated below the thumbnail in a searchable, screen reader-ready format.
- Extended text description of the important visuals such as graphs and charts presented in the slides.
- Any tabular data from the video is recreated and properly labeled for screen reader navigation and reading.
- All math equations are presented in MathML that provides both content and presentation if on screen.
- Transcript that captures all of the original speech in the video labeled by the person speaking.

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Module 3 Inferential and Predictive Statistics for Business

Lesson 3-1 An Introduction

Lesson 3-1.1 An Introduction

[Media Player for Video](#)

Examples - Slide 1

- Does eating nuts help you live longer?
- Does Zika virus cause a serious birth defect?
- Does regular exercise improve memory?
- Does burning fossil fuels accelerate global warming?

Transcript

Does eating nuts help you live longer? Does Zika virus causes a serious birth defect? Does regular exercise improve memory? Does burning fossil fuels accelerate global warming? We can use regression to answer questions like these. In the last two lessons, we focused on building confidence intervals and comparing different population. That is what we call inferential statistics, using sample information to get some insight about the population. Now, we can go one step further; think about it, look at the first question on this slide. From what we have learned so far, I know I can do a study comparing people who eat nuts to those who don't, and then find whether the mean life expectancy of the two groups are different or not. While this is an insight worth having, you may face the next set of questions that may come up, like how much of an increase is due to eating nuts? Maybe one group was healthier than the other group to begin with; or how much nuts does one need to eat to get this benefit? In another word, do you have a way of explaining how consuming nuts impacts longevity among all other factors that could be impacting longevity? We can use regression to find the answer, and once we know how nut consumption and longevity are related, now we can go ahead and start predicting. For example, we can say, eating 100 grams per day will reduce the risk of dying, by the way, this is a result of a real study. Regression builds on everything we have learn so far in the class and would allow us to move to prediction based on sample information.

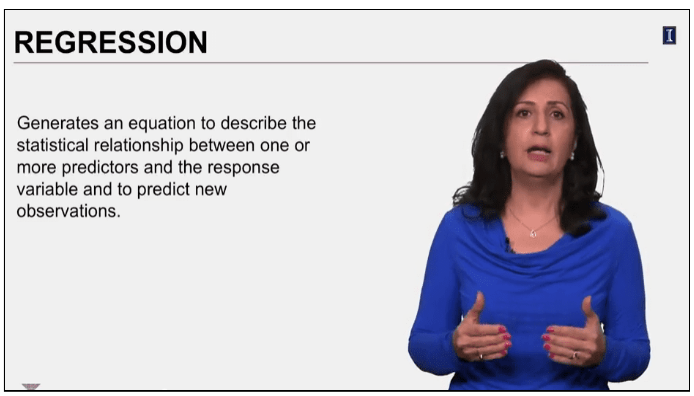
Objectives of Regression Analysis - Slide 2

- Prediction – i.e., making inferences about possible cause-effect relationships and extrapolating them into the future
- Address questions of why so much or why so many
- Dynamic – used to make process improvements and prescribe better ways of doing work
- Expect future behavior of process to be similar to past

Transcript

The main objectives of regression analysis is first, the main is prediction, like making inferences about possible cause and effect relationships and extrapolate them into future; then, gaining the ability to address questions of why so much or why so many cause this difference, taking the new understanding and improving our business. One caveat here is the fact that we expect future behavior of the process to be similar to the past. At some time in the future, the relationship that we have discovered may weaken or cease to exist, so watch how well the predictions are turning out and adjust the model when needed.

Regression - Slide 3



Generates an equation to describe the statistical relationship between one or more predictors and the response variable and to predict new observations.

Transcript

Regression analysis will generate an equation to describe the statistical relationship between one or more predictors and the response variable, and then we can use the regression equation to predict the outcome of new observations. In the field of statistic, regression is a big topic on its own and there are many models available for regression analysis. The simplest model is the simple linear regression and that is what I will introduce you to in this module.

Simple Linear Regression - Slide 4

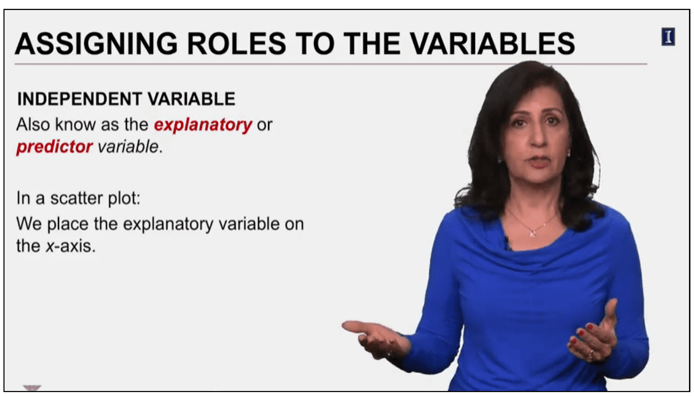
Two variables:

I. Dependent
II. Independent

Transcript

There are two different variables that make up a simple linear regression model, the dependent variable and independent variable. The independent variable, which is also known as the explanatory or predictor variable, this is the variable we are going to use to try and understand how it impacts the dependent variable. In a business example, this might be advertising level or time period, we can control the independent variable, for example, one can decide how much to spend on advertising.

Assigning Roles to the Variables (1 of 2) - Slide 5



Independent Variable

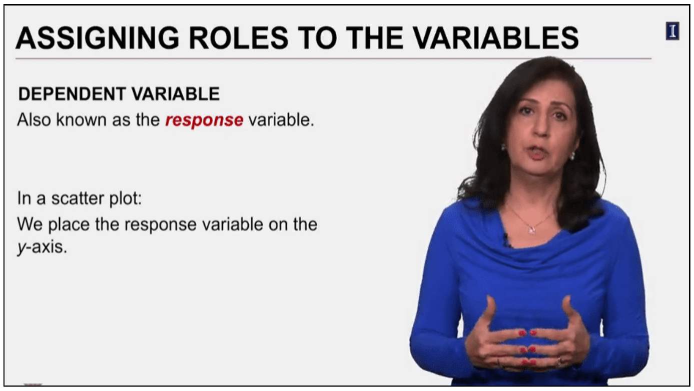
Also known as the ***explanatory*** or ***predictor*** variable.

In a scatter plot: We place the explanatory variable on the x-axis.

Transcript

Simple linear regression most often starts with a scatter plot, which will allow us to visually see if we see a relationship exist between the two variables, and whether or not the relationship has a linear form. For the scatter plot, we place the explanatory variable on the x-axis.

Assigning Roles to the Variables (2 of 2) - Slide 6



Dependent Variable Also known as the ***response*** variable.

In a scatter plot: We place the response variable on the y-axis.

Transcript

The dependent variable is also known as the response variable, it is the variable that we wish to understand or predict; in a business example this might be sales or demand. For the scatter plot, we place the response variable on the y-axis. A regression model is considered a simple linear regression, when there is only one dependent variable and one independent variable, and the relationship between these two has a linear form. You have to be very careful not to confuse the two.

Which is the Predictor? (1 of 2) - Slide 7

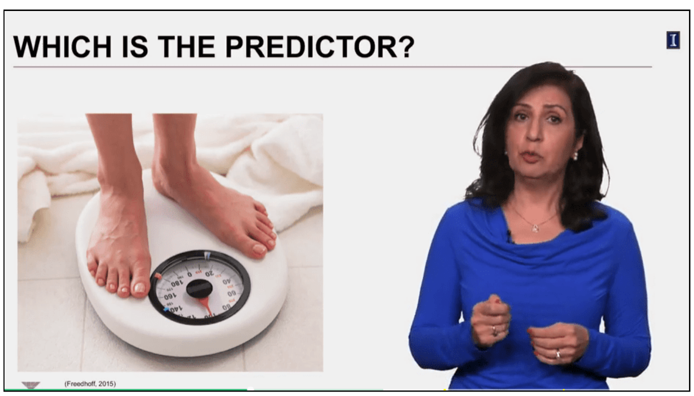


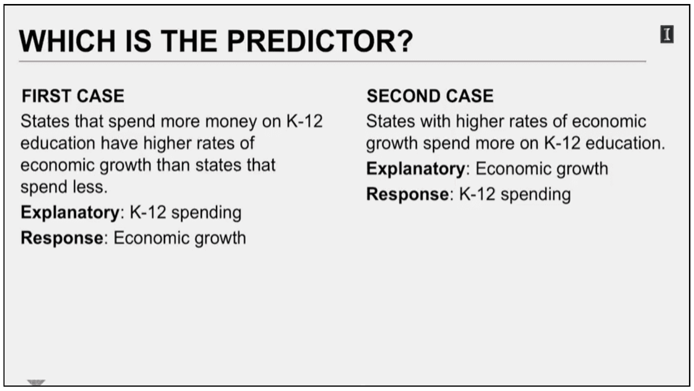
Image of a person standing on a scale

(Freedhoff, 2015)

Transcript

When you step on a weight scale, do you ever wonder that according to the rate chart, you have a height problem? You should be seven feet tall for the number you see on your scale? Of course not. We understand the height is used to estimate our weight, so height is the predictor and weight is the response variable. While in this example, rules variables play may be easy to see, there are many times that we confuse the two.

Which is the Predictor? (2 of 2) - Slide 8



First Case

States that spend more money on K-12 education have higher rates of economic growth than states that spend less.

Explanatory: K-12 spending

Response: Economic growth

Second Case

States with higher rates of economic growth spend more on K-12 education.

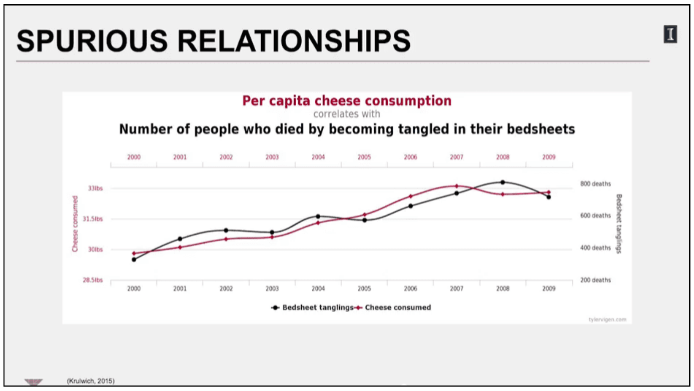
Explanatory: Economic growth

Response: K-12 spending

Transcript

It is not always easy to detect which variable is the explanatory variable and which variable is the response variable. Imagine a study that wants to establish a connection between K-12 spending and economic growth. If you get a report that says, "States that spend more money in K-12 education have higher rates of economic growth than states that spend less", then the study is making K-12 spending as the explanatory variable, and the state's rate of economic growth as the prediction variable; meaning, investing in K-12 causes economic growth. On the other hand, I could argue that states that have a good rate of growth have more money and thus spend more on K-12. In this understanding, the economic growth rate is causing how much money is spent on K-12. What you see here is an example where the direction of cause and effect it's not that clear. The two variables are hopelessly tangled, since they both can affect one another at different times. In the first case, the struggling state decides to increase investment in K-12 so that its residents are ready to be employed, and that sometime in the future when this investment pays off, receive the second case occurring, which is since the state is doing really well and has money, it will start spending more on K-12, does the variables switch roles? Avoid these types of studies. When you pick your variables, you should have reason to believe that the explanatory variables affect the dependent variable and not the other way around. Just because a strong relationship between two variables exist, doesn't necessarily mean that they are functionally related. Any two sequences y and x, that are monotonically related, if x increases, then y either increases or decreases, would always show a strong statistically relationship, while functional relationship may not exist.

Spurious Relationships - Slide 9



A line graph entitled, Per capita cheese consumption correlates with number of people who died by becoming tangled in their bedsheets. The x-axis is in years and ranges from 2000 to 2009. There are 2 y-axes: the leftmost y-axis denotes cheese consumption and ranges from 28.5 pounds to 33.5 pounds; the rightmost y-axis denotes bedsheet tanglings and ranges from 200 deaths to 1000 deaths. The two lines on the graph, one denoting bedsheet tanglings and the other cheese consumption, follow a very similar trend. The data in tabular form are as follows:

Cheese consumption and number of bedsheet tanglings by year

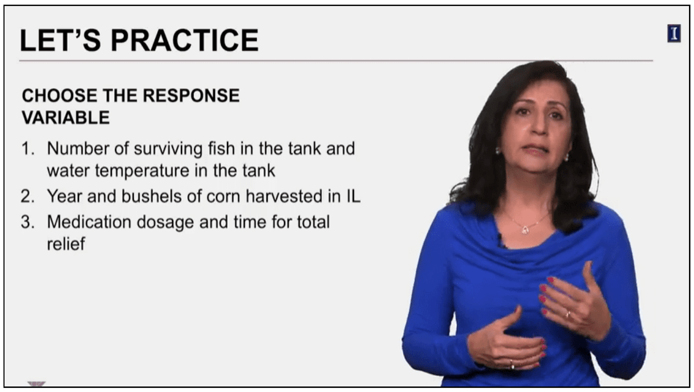
Year	Bedsheet tanglings (deaths)	Cheese consumed (lbs.)
2000	320	29.8
2001	480	30.15
2002	520	30.5
2003	500	30.45
2004	605	31.25
2005	600	31.7
2006	680	32.5
2007	780	33.15
2008	820	32.75
2009	760	32.85

(Krulwich, 2015)

Transcript

Look at this graph, which shows cheese consumption is highly correlated to a number of people who died by becoming tangled in their bedsheets, no kidding, data is from National Geographic. So, make sure that you pick variables that are functionally related.

Let's Practice - Slide 10



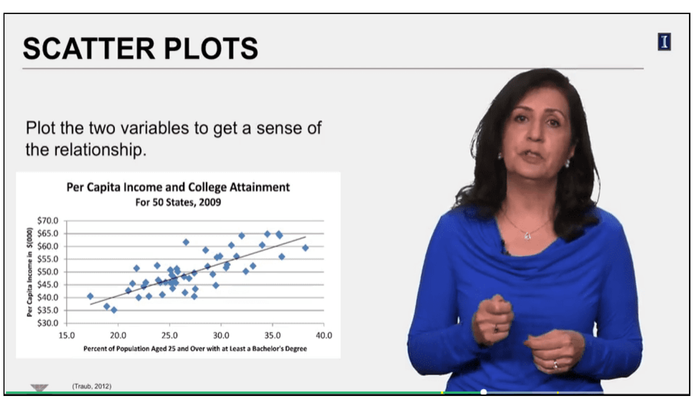
Choose the response variable

1. Number of surviving fish in the tank and water temperature in the tank
2. Year and bushels of corn harvested in IL
3. Medication dosage and time for total relief

Transcript

So, now let's practice. Imagine conducting these studies. In one study, we are collecting data on number of surviving fish in the tank and water temperature in the tank; in the second study, we are recording the year and the bushels of corn harvested in Illinois and in the third study, it's collecting data on medication doses and time elapsed for total relief. Identify for each study the response variable. The response variable in each study is shown in red color font. In the first study, the number of surviving fish in the tank is responding to the water temperature in the tank; in the second study, bushels of corn harvested in Illinois is being tracked year after year so the harvest is the response variable; and in the third study, time elapsed for total relief is the response variable and depends on medication dosage.

Scatter Plots - Slide 11



Plot the two variables to get a sense of the relationship.

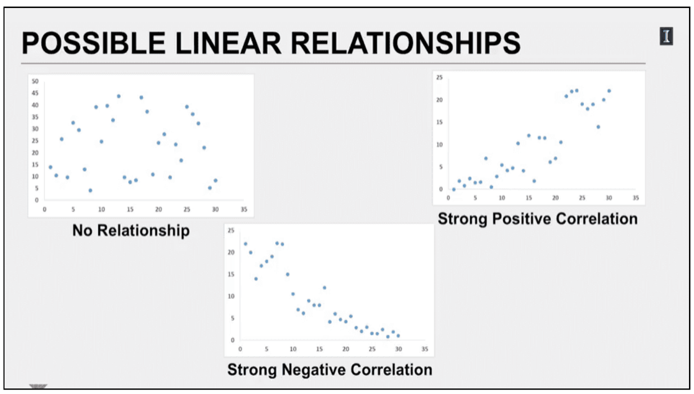
A scatter plot entitled, Per capita income and College attainment, for 50 States in 2009. The x-axis is the percent of population aged 25 and over with at least a bachelor's degree, ranging from 15 to 40. The y-axis is per capita income in $(000), ranging from 30 to 70. The plot has an almost linear increase trend.

(Traub, 2012)

Transcript

We learned about scatter plots early on. This is a great visual tool for detecting whether or not the variables we have identified show any special relationship with one another. For instance, here's a scatter plot for income versus educational attainment for all 50 states. Looking at the scatter plot where each diamond represents a state, we can see that there is a linear relationship between the predictive variable, educational level on the x-axis, and income, the response variable on the y-axis. Since as the education level goes up, so does the income, then will we conclude that these two variables are positively correlated; we can even look at the spread of the points and see that they are fairly clustered, which suggests a strong relationship. So, it is well worth doing the full analysis to know more in detail by how much does education level impacts the income level.

Possible Linear Relationships - Slide 12

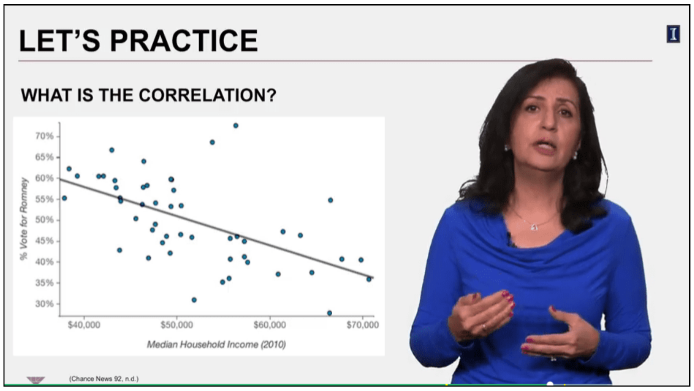


Three scatter plots. The upper left one is a scatter plot of no relationship with the data points looking randomly from everywhere. The upper right one is a scatter plot of strong positive correlation with the increasing of x, the data points' corresponding y also increase, which forms a linear increase trend. The last plot represents the strong negative correlation with the increasing of x-value, the data points' corresponding y value decrease, which forms a decreasing linear trend.

Transcript

Plotting the two variables, we can detect whether we have identified two variables, which don't have any correlation at all, in which case you don't need to pursuit building a predictive model, or we can detect a positive correlation or a negative correlation. These two will be suitable for simple linear regression analysis. So, again, let's practice.

Let's Practice - Slide 13



What is the correlation?

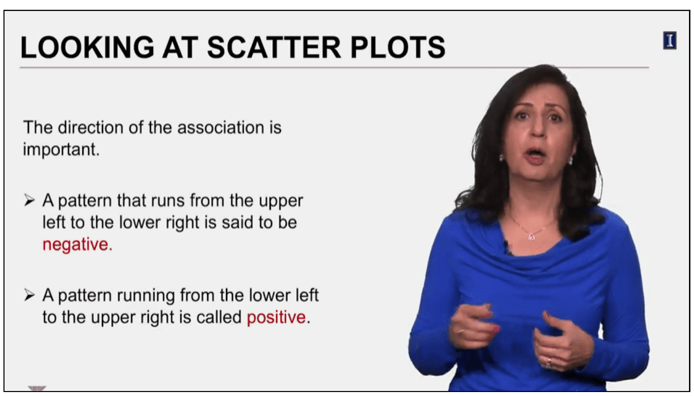
A scatter plot with the x-axis as the median household income from $40,000 to $70,000 and the y-axis as the percentage vote for Romney from 30% to 70%. With the increase of median income, the vote percentage decreases.

(Chance News 92, n.d.)

Transcript

This scatter plot shows the correlation between per capita income in a state and the percent of the 2012 presidential election votes that went to Romney. What type of correlation do you see here?

Looking at Scatter Plots (1 of 4) - Slide 14



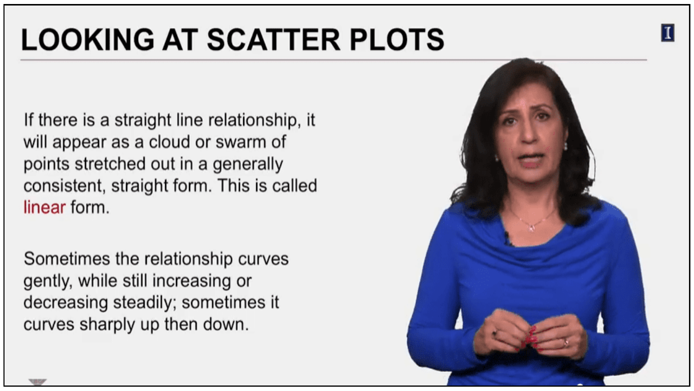
The direction of the association is important

- A pattern that runs from the upper left to the lower right is said to be **negative**.
- A pattern running from the lower left to the upper right is called **positive**.

Transcript

States medium income is negatively correlated with proportion of the state's votes in the 2012 election that went to Romney. When you have a scatter plot, first thing to look for in a scatter plot is the direction of the association, a pattern that runs from the upper left to the lower right is set to be negative, a pattern that running from the lower left to the upper right is called positive.

Looking at Scatter Plots (2 of 4) - Slide 15



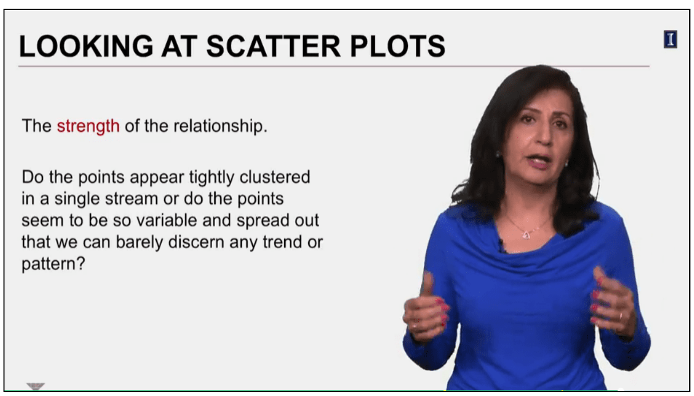
If there is a straight-line relationship, it will appear as a cloud or swarm of points stretched out in a generally consistent, straight form. This is called **linear** form.

Sometimes the relationship curves gently, while still increasing or decreasing steadily; sometimes it curves sharply up then down.

Transcript

The second thing to look for in a scatter plot is its form. If there's a straight-line relationship it will appear as a cloud or swarm of points stretched out in a generally consistent, straight form, this is called linear form. Sometimes the relationship curves gently, while still increasing or decreasing steadily; sometimes it curves sharply up and down. If you see this pattern, then a linear regression model will not be suitable.

Looking at Scatter Plots (3 of 4) - Slide 16



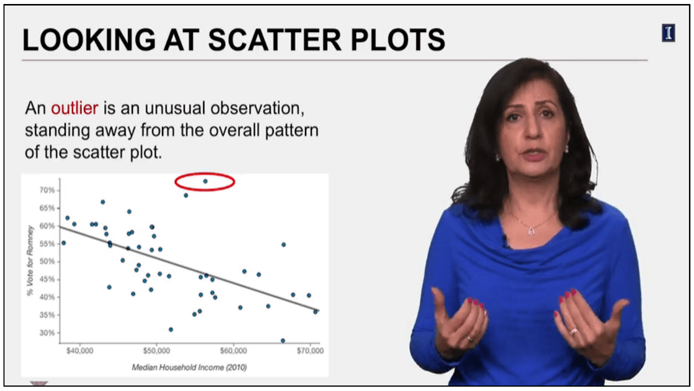
The **strength** of the relationship.

Do the points appear tightly clustered in a single stream or do the points seem to be so variable and spread out that we can barely discern any trend or pattern?

Transcript

Third feature to look for in a scatter plot is the strength of the relationship. Do the points appear tightly clustered in a single stream or do the points seem to be so variable and spread out that that we can barely discern any trend or pattern? As the points get more and more spread out, then the relationship becomes weaker and weaker. A good model will have most of its points closely clustered, providing a strong relationship.

Looking at Scatter Plots (4 of 4) - Slide 17



An **outlier** is an unusual observation, standing away from the overall pattern of the scatter plot.

This slide shows the same plot as [Slide 13 Let's Practice](). Here, the highest data point at around $56,000 with 73% of vote is highlighted representing an outlier in the graph.

Transcript

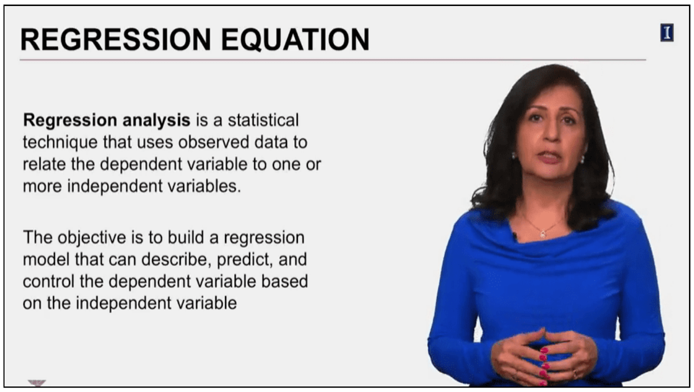
Finally, always look for the unexpected; an outlier is an unusual observation, standing away from the overall pattern of the scatter plot. Here's the graphic shows income versus percent of votes for Romney. In this graph, this point can be defined as an outlier. Sometimes we remove the outlier from the data before doing the regression, but other times we may want to know why the outlier is there before deciding what to do next. Scatter plots are useful in visualizing the relationship between the two variables we're studying. Once we have a sense that the relationship exists, then we can move to the next step which involves defining your mathematical equation, which describes the relationship between these two variables. This will be called "the regression equation".

Lesson 3-2 Least Square Methods

Lesson 3-2.1 Least Square Methods

[Media Player for Video](#)

Regression Equation (1 of 6) - Slide 18



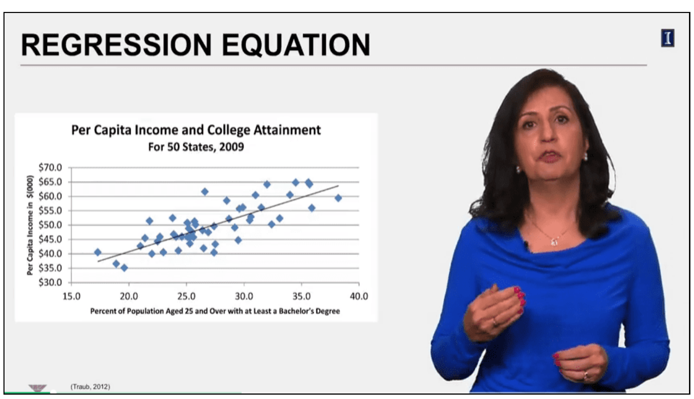
Regression analysis is a statistical technique that uses observed data to relate the dependent variable to one or more independent variables.

The objective is to build a regression model that can describe, predict, and control the dependent variable based on the independent variable

Transcript

Regression analysis is a statistical technique that uses observed data to relate the dependent variable to one or more independent variables. The objective of regression analysis is to build a regression model that can describe, predict, and control the dependent variable based on the independent variable.

Regression Equation (2 of 6) - Slide 19



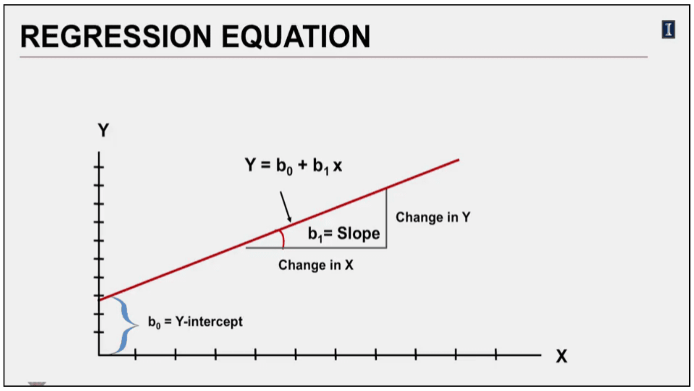
This slide shows the same plot as [Slide 11 Scatter Plots](#).

(Traub, 2012)

Transcript

Going back to the scatter plot, for level of education and per capita income for the fifty states of the U.S., the black line that goes through the individual data points, the blue diamonds, is drawn based on regression equation. This line and its equation is calculated based on a method known as the least square method. Least square method will develop linear equation which relates y-variable to the x-variable.

Regression Equation (3 of 6) - Slide 20

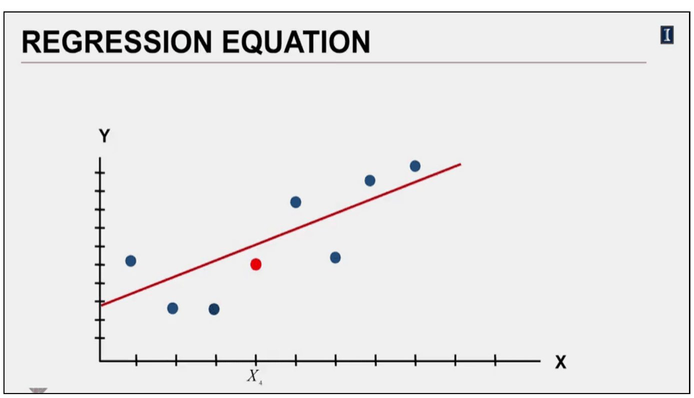


There is a graph with a positive line with the equation $Y = b_0 + b_1 x$ in the x-y axis. When x = 0, the intercept of the line is the correspond y-value for x = 0, which is b_0. For slope b_1, it stands for the change in y when you change 1 unit of x. This can be illustrated with a right triangle where the leg is y, the base-side is x and the corresponding angle is b_1.

Transcript

This equation is defined by where it intersects the y-axis, this is known as the b0 and its slope, which is calculated by taking the rate of change in y by the rate of change in x and is the slope of the line, and is represented by b1. Once we have the regression equation, then for a given x-value, we can solve and predict the value for variable y. However, they're not going to be exactly right in our prediction, you will have some error.

Regression Equation (4 of 6) - Slide 21

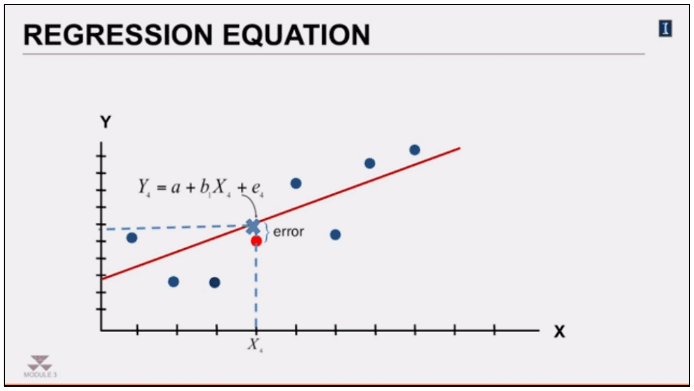


Scatter plot with 8 data points with increase correlation in x-y axis. 7 of them are marked blue and the fourth one, from left to right, with x value = x_4 is marked red. There is also a red increasing regression line. Half of the data points are above the line, while the other half are below it.

Transcript

For example, for this point here, which is the 4th value for x, if we use the equation, the predicted value of y will be here.

Regression Equation (5 of 6) - Slide 22

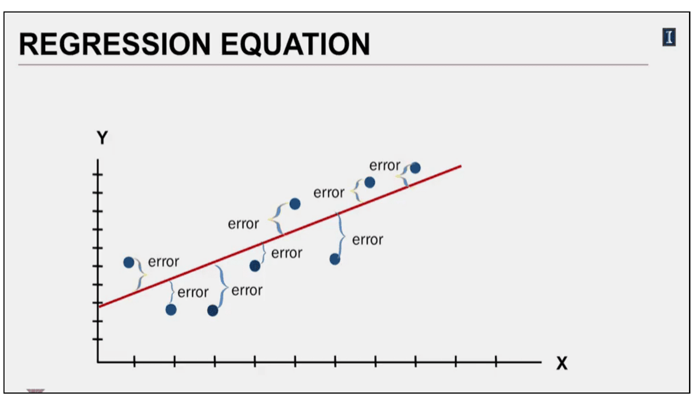


This slide shows the same plot as Slide 21 Regression Equation (4 of 6). Here, the distance from the red point where $x = x_4$ to the regression line is labeled as the error. In addition, the following formula: $Y_4 = a + b_1X_4 + e_4$

Transcript

The difference between our predicted value and the actual observed value is what we call Prediction Error or just error or the residual value. So, including this error, the prediction value will look like this. So now, think of fitting the line to all these points. For the data we have, we can actually calculate the error for each point and its estimated value. For points that are below the line, our prediction will be a higher value than we observed, and for points that are above the line, our prediction will be a lower value than what we observed.

Regression Equation (6 of 6) - Slide 23

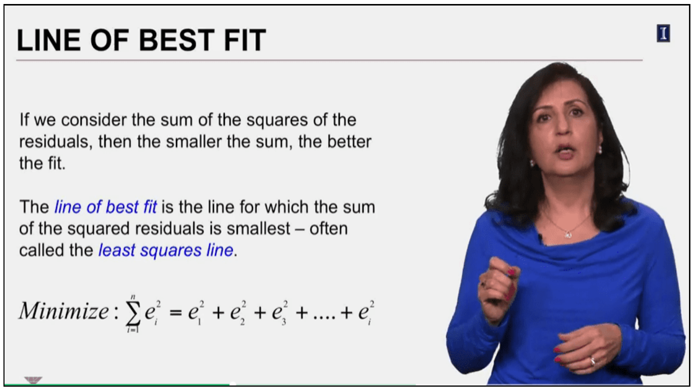


This slide shows the same plot as Slide 21 Regression Equation (4 of 6). Here, the error for every data points is drawn in the picture. The vertical distance between each data point and the regression line is the error for each data point.

Transcript

As you saw, some residuals will be positive and some negative. So, adding them all up is not a good assessment of how well the line fits the data. If you consider sum of the squares of the residuals, then the smaller the sum, the better the fit.

Line of Best Fit - Slide 24



If we consider the sum of the squares of the residuals, then the smaller the sum, the better the fit.

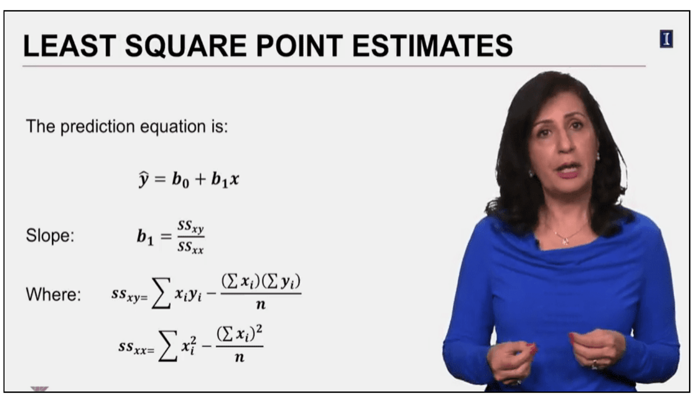
The ***line of best fit*** is the line for which the sum of the squared residuals is smallest – often called the ***least squares line***.

$$\text{Minimize}: \ \sum_{i=1}^{n} e_i^2 = e_1^2 + e_2^2 + e_3^2 + ... + e_n^2$$

Transcript

The line of best fit is the line for which the sum of the squared residuals is smallest, this is what is often called the least square line, using the least square method.

Least Square Point Estimates (1 of 2) - Slide 25



The prediction equation is : $\hat{y} = b_0 + b_1 x$

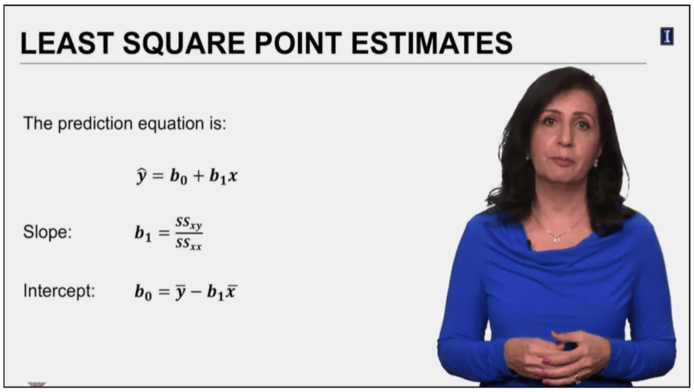
Slope : $b_1 = \frac{SS_{xy}}{SS_{xx}}$

Where : $SS_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right) \times \left(\sum_{i=1}^{n} y_i\right)}{n}$ and $SS_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$

Transcript

Then the estimated value y, as predicted by the regression equation, is we can add ŷ equals b sub 0 plus b sub 1 times x. I will probably drop the hat in most of my slides, but technically, the hat over the y is a reminder that we are dealing with prediction. But that is all we will be doing here anyway, so just let's remember and make life a little less cluttered. B1, the slope is based on two sums of squares and these sums of squares are calculated using the following formulas. Now, I can guess what you're thinking. These formulas look less than user friendly. As before, we will be doing the analysis using Excel. So, most everything is automated. But to understand the output, it is important to know how they are being derived. So, be patient, and stay with me. Soon, it will all come together.

Least Square Point Estimates (2 of 2) - Slide 26



The prediction equation is : $\hat{y} = b_0 + b_1 x$

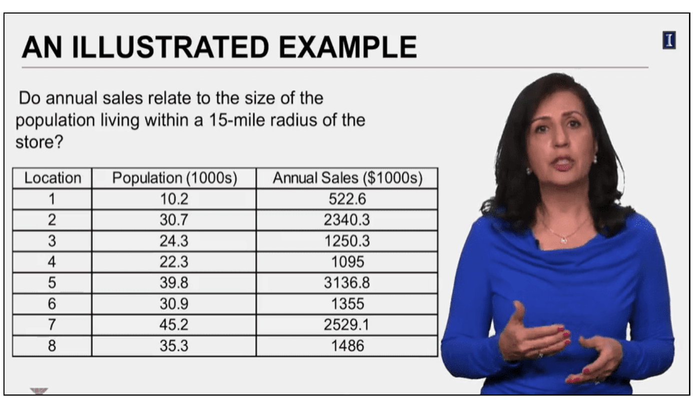
Slope : $b_1 = \dfrac{SS_{xy}}{SS_{xx}}$

Intercept : $b_0 = \bar{y} - b_1 \bar{x}$

Transcript

Once the slope value of b sub 1 has been calculated we can calculate b sub 0, the y intercept, by taking the average value of y minus the slope times the average value of axis.

An Illustrated Example - Slide 27



Do annual sales relate to the size of the population living within a 15-mile radius of the store?

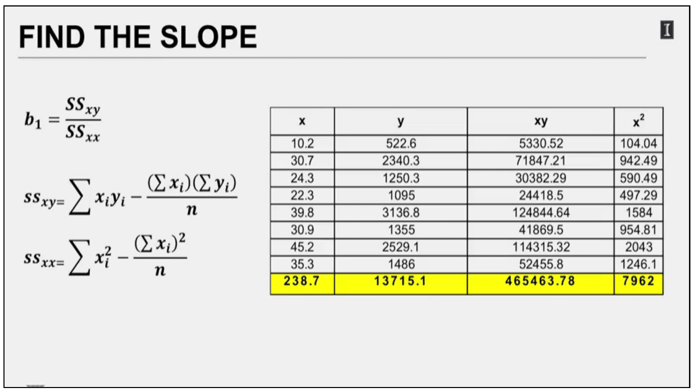
Sales by location to the store

Location	Population (1000s)	Annual Sales ($1000s)
1	10.2	522.6
2	30.7	2340.3
3	24.3	1250.3
4	22.3	1095
5	39.8	3136.8
6	30.9	1355
7	45.2	2529.1
8	35.3	1486

Transcript

Imagine having eight different stores at different locations. You are wondering if the annual sales at a store is dependent on how many people live within a 15-mile radius of the store. If you could establish that, in fact, this is the case, then in future when we want to open up a new store, we can look at alternative sites and predict the annual sales for each possible location and then pick the one with the highest predicted sales. Let me take you to how we would solve this. First, I will use the equations, and later show you how Excel will present the same results to you. Again, my objective here is that by going to the math, Excel often will make more sense.

Find the Slope (1 of 2) - Slide 28



FIND THE SLOPE

$$b_1 = \frac{SS_{xy}}{SS_{xx}}$$

$$ss_{xy=} \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

$$ss_{xx=} \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

x	y	xy	x^2
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	13715.1	465463.78	7962

$$b_1 = \frac{SS_{xy}}{SS_{xx}}$$

$$SS_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right) \times \left(\sum_{i=1}^{n} y_i\right)}{n}$$

$$SS_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

Slope examples

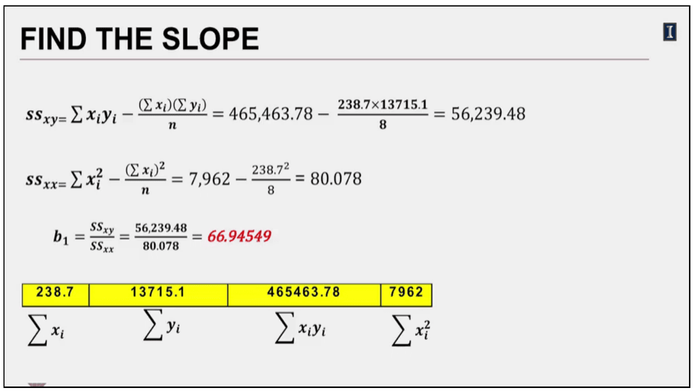
x	y	xy	x^2
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

In the table, the row where x = 238.7 is highlighted.

Transcript

These are the equations I will be using. First, I take the data we have and calculate the numbers I need. So, for example, 238.7 is the sum of all values in the column with the heading of x. Once I do this for all columns, I would have everything I need to find the slope and these values are in the row highlighted in yellow. And each number is one of the notations used in the slope calculations.

Find the Slope (2 of 2) - Slide 29



$$SS_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right) \times \left(\sum_{i=1}^{n} y_i\right)}{n} = 465463.78 - \frac{238.7 \times 13715.1}{8} = 56239.48$$

$$SS_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 7{,}962 - \frac{238.7^2}{8} = 80.078$$

$$b_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{56239.48}{80.078} = 66.94549$$

$$\sum_{i=1}^{n} x_i = 238.7$$

$$\sum_{i=1}^{n} y_i = 13715.1$$

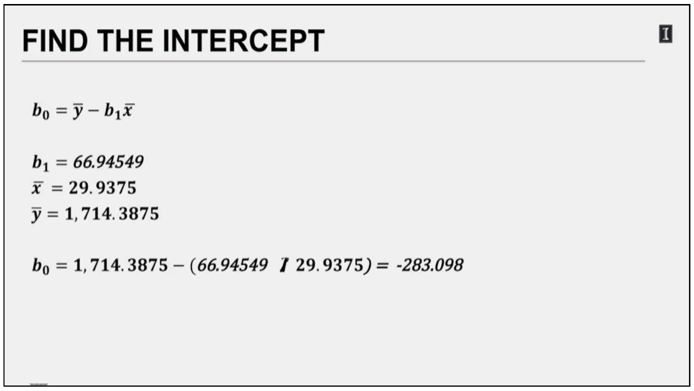
$$\sum_{i=1}^{n} x_i y_i = 465463.78$$

$$\sum_{i=1}^{n} x_i^2 = 7962$$

Transcript

Using the summary values, first find ss of xy. Ss stands for sum of the squares which we calculate here. Then use the numbers to calculate ss of xx. Now, we take the ratio of the two to find the slope. This says for every increase in value of x, the annual sale will go up by $66.94, slope is the rate of change.

Find the Intercept - Slide 30



$b_0 = \bar{y} - b_1\bar{x}$

$b_1 = 66.94549$

$\bar{x} = 29.9375$

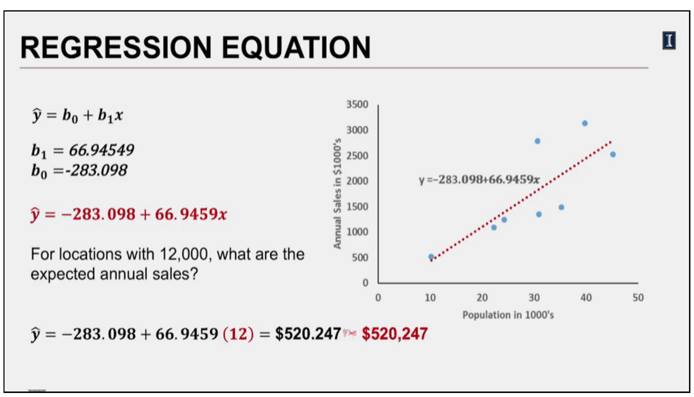
$\bar{y} = 1714.3875$

$b_0 = 1714.3875 - (66.94549 \times 29.9375) = -283.098$

Transcript

Now, we can calculate the intercept b sub 0. Again, substituting numbers for the notation, we can find the intercept to be negative 283.098.

Regression Equation - Slide 31



$\hat{y} = b_0 + b_1x$

$b_1 = 66.94549$

$b_0 = -283.098$

$\hat{y} = -238.98 + 66.9459x$

For locations with 12,000, what are the expected annual sales?

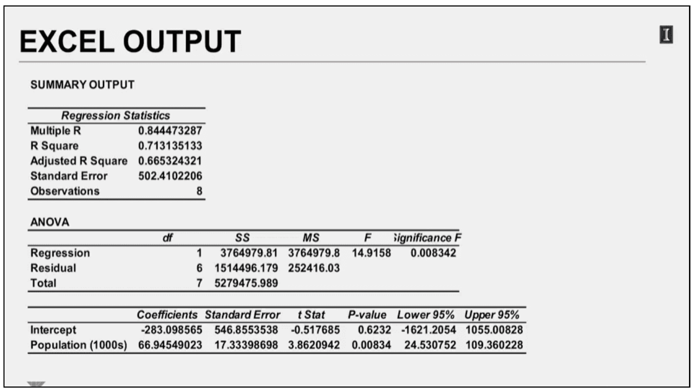
$\hat{y} = -238.98 + 66.9459 \times (12) = \$520.247$

Scatter plot where the x-axis is the population in 1000s from 0 to 50 and the y-axis is the Annual sells in $1,000’s from 0 to 3500. In the plot there is a regression line with the equation: $y = -283.098 + 66.9459x$.

Transcript

Now, we can write the regression equation. This is the complete regression equation. Plotting our data, the scatter plot and the regression line will look like this. The intercept is a negative value. Just imagine extending the line until it will intercept the y. That would yield a negative value and the slope of the line is 66.94549. So, this means when you increase x by 1 and here 1 will represent thousands since the numbers were scaled. The annual sale will increase by about 67 which is really $67,000. Now, for location, the 12,000 nearby, then what is the estimated average annual sales? We will put 12 in x, again numbers are scaled by thousands, and the answer will be $520,247.

Excel Output (1 of 3) - Slide 32


EXCEL OUTPUT

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.844473287
R Square	0.713135133
Adjusted R Square	0.665324321
Standard Error	502.4102206
Observations	8

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3764979.81	3764979.8	14.9158	0.008342
Residual	6	1514496.179	252416.03		
Total	7	5279475.989			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-283.098565	546.8553538	-0.517685	0.6232	-1621.2054	1055.00828
Population (1000s)	66.94549023	17.33398698	3.8620942	0.00834	24.530752	109.360228

Summary Output

Regression Statistics

Multiple R	0.844473287
R Square	0.713135133
Adjusted R Square	0.665324321
Standard error	502.4102206
Observations	8

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3764979.81	3764979.8	14.9158	0.008342
Residual	6	1514496.179	252416.03		
Total	7	5279475.989			

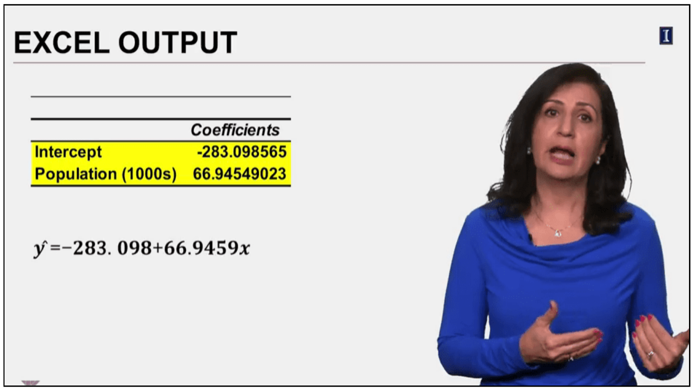
Other statistics

	Coefficients	Standard error	t-stat	p-value	Lower 95%	Upper 95%
Intercept	−283.098565	546.8553538	−0.517685	0.6232	−1621.2054	1055.00828
Population (1000s)	66.94549023	17.333986698	3.8620942	0.00834	24.530752	109.360228

Transcript

Now look at what we get if we use Excel to solve this problem for us. Excel returns three parts in this analysis. Right now, I want you to focus on the last table, this part.

Excel Output (2 of 3) - Slide 33



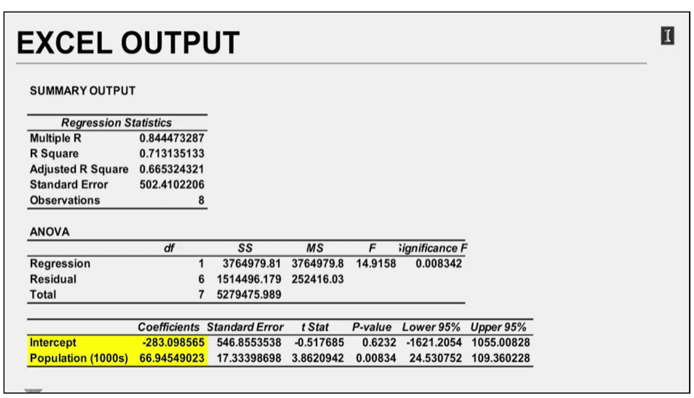
The slide shows the same table titled, Other statistics, as in Slide 32 Excel Output (1 of 3), but with just the coefficients column.

$\bar{y} = -283.098 + 66.9459x$

Transcript

Focusing on the last table, the highlighted values are showing the coefficients for the intercept be 0 and the coefficient for the random variable population. That's our b1. Now look at what we got when we did the calculations using the equation. Now, surprisingly, it is the same. There is not more to do that just finding the equation and regression analysis.

Excel Output (3 of 3) - Slide 34



The slide shows the same tables as Slide 32 Excel Output (1 of 3). Here, the coefficients of the Intercept and the Population are highlighted.

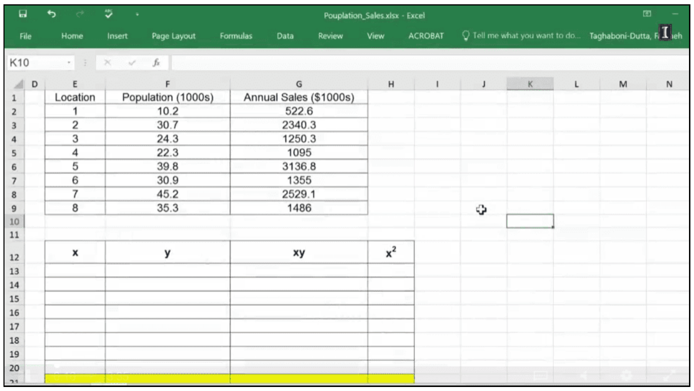
Transcript

This is why the output has so much more information than the intercept and the slope of the regression equation. We will be learning about all these in the remaining lessons in this module.

Lesson 3-2.2 Least Square Method in Excel

Media Player for Video

Population Sales Excel Worksheet (1 of 8) - Slide 35



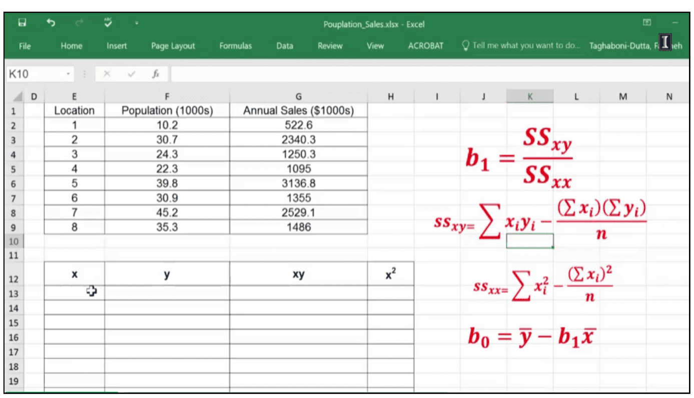
The slide shows the Population sales data set with two tables. The first table has three columns: Location, Population (1000s), and Annual sales ($1000s). The second table has four columns: x, y, xy, and x^2. This second table has no values in it.

Download the Square Method Excel file (Refer to Data - First worksheet)

Transcript

This is an example that I have shown you in the PowerPoint, and I'm going to now show you how I calculated the least squares method. As I have mentioned, I don't expect you to do this on your own, because we will use the regression function, within the data analysis. It's good for you to know where the values and the output is coming from, mainly this slope and the intercept for the regression line.

Population Sales Excel Worksheet (2 of 8) - Slide 36



The slide shows the following formulas:

$$b_1 = \frac{SS_{xy}}{SS_{xx}}$$

$$SS_{xy} = \sum_{\mathrm{i}=1}^{n} x_i y_i - \frac{\left(\sum_{\mathrm{i}=1}^{n} x_i\right) \times \left(\sum_{\mathrm{i}=1}^{n} y_i\right)}{n}$$

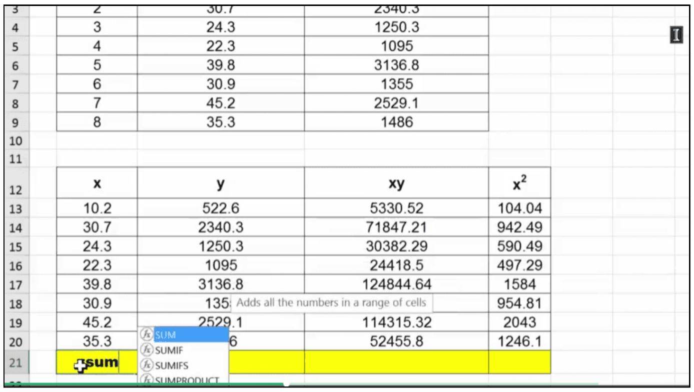
$$SS_{xx} = \sum_{\mathrm{i}=1}^{n} x_i^2 - \frac{\left(\sum_{\mathrm{i}=1}^{n} x_i\right)^2}{n}$$

$b_0 = \bar{y} - b_1\bar{x}$

Transcript

So, I'm going to show you with the formulas, that you see how I can calculate the slope, as well as the intercept. So, I'm going to copy my x-values down below, and I'm going to do the same thing for the y-values.

Population Sales Excel Worksheet (3 of 8) - Slide 37



3	2	30.7	2340.3
4	3	24.3	1250.3
5	4	22.3	1095
6	5	39.8	3136.8
7	6	30.9	1355
8	7	45.2	2529.1
9	8	35.3	1486
10			
11			

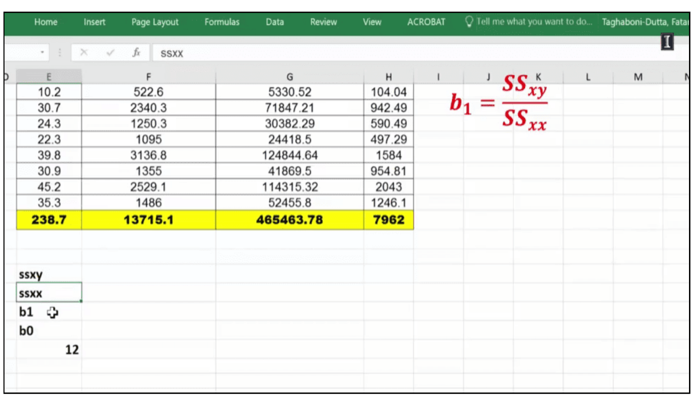
12	x	y	xy	x^2
13	10.2	522.6	5330.52	104.04
14	30.7	2340.3	71847.21	942.49
15	24.3	1250.3	30382.29	590.49
16	22.3	1095	24418.5	497.29
17	39.8	3136.8	124844.64	1584
18	30.9	135		954.81
19	45.2	2529.1	114315.32	2043
20	35.3	6	52455.8	1246.1
21	=sum			

The slide shows the tables from [Slide 35 Population Sales Excel Worksheet (1 of 8)](), except the second table is filled out and the last row is highlighted. In the last row of the column titled, x, =SUM is written to calculate the sum of this column.

Transcript

So, x is my population, and it's being used to predict the y, which is the annual sales in the store. So, this column is going to be in x times y, so, it's simply this cell multiplied by this cell. And, you really don't need to repeat this, over and over again. And, I'm going to just use the cross-hair again and double-click because, even though I can drag this, if I had thousands of data, that's not really the most efficient way, so I'm just going to do this, and it will populate everything for you. And then, the next column is just x square, which is just this value raised to the power of 2. So, once again, I can just do this, and I will get all the values populated here. Now that I have this, I can calculate the sum for every column, so, right here is going to be the sum of x's, and once I have this, I can drag this, and I will get values for sum of y's, sum of x's times y's, and sum of x squares.

Population Sales Excel Worksheet (4 of 8) - Slide 38



E	F	G	H
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

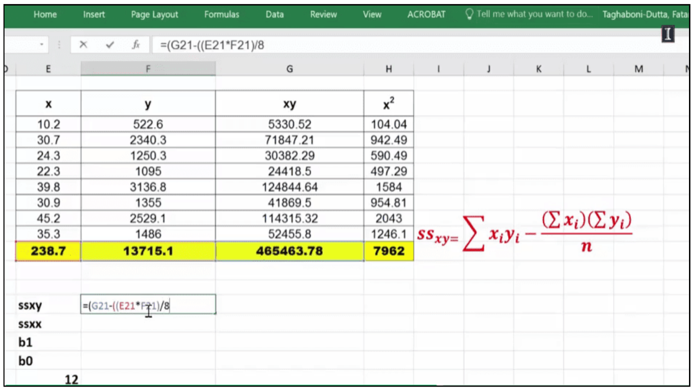
ssxy	
ssxx	
b1	
b0	
	12

The slide shows the outcome of each of the columns: x column is 238.7, y column is 13715.1, xy column is 465463.78, and x^2 column is 7962. Below the table, four new cells called "ssxy", "ssxx", b_1, and b_0 are created and vertically listed.

Transcript

So, as you can see for the equation for b1, I would need to have the value of SS xy and SS of xx, these are sum of squares, in order for me to take their ratio and find b1.

Population Sales Excel Worksheet (5 of 8) - Slide 39



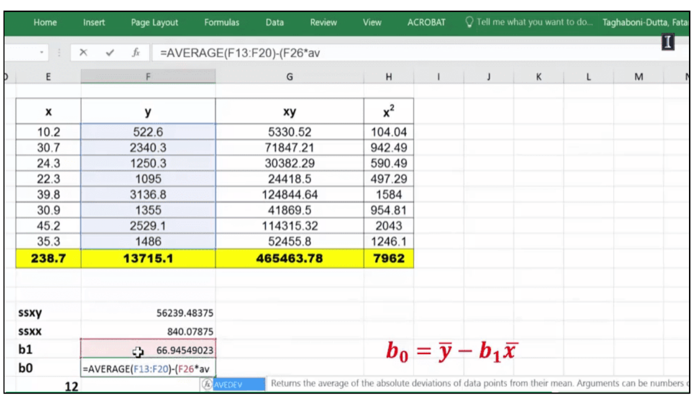
x	y	xy	x^2
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

The slide shows how to calculate the ssxy value. The function used is =(G21-((E21*F21)/8)), where G21 is 465463.78 (xy sum), E21 is 238.7 (x sum), and F21 is 13715.1 (y sum).

Transcript

The equation for sum of squares of x's times y's, which is SS xy, is equal to parenthesis, sum of squares of xy's, this value, minus sum of x's multiplied by sum of y's, divided by n, and in this case, n is 1, 2, 3, 4, 5, 6, 7, 8, we have 8 stores, and this will be the value for SS of xy.

Population Sales Excel Worksheet (6 of 8) - Slide 40



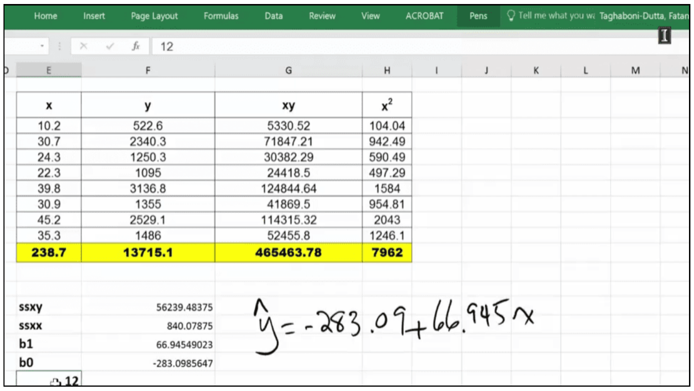
x	y	xy	x^2
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

The slide shows the outcome of ssxy, which is 56239.48375. In addition, the ssxx value is 840.07875 and the b1 value is 66.94549023. Here, the slide intends to calculate the b0 value. Th function of this value is =AVERAGE(F13:F20)-(F26*AVEDEV), where F13 to F20 are all the values of the y column, except for the summation, and F26 is 66.94549023 (b1 value).

Transcript

Then, SS of xx is, as you can see in the equation, sum of x squares, so that's this value, minus sum of x's, that are now raised to the power of 2, divided by 8. Now, b1 is simply, as you can see in the equation, this value divided by this value. And, b0 is the average of y values. So, I'm going to take the average here, minus b1, which we just calculated, times the average of, and that will be our equation.

Population Sales Excel Worksheet (7 of 8) - Slide 41



x	y	xy	x^2
10.2	522.6	5330.52	104.04
30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

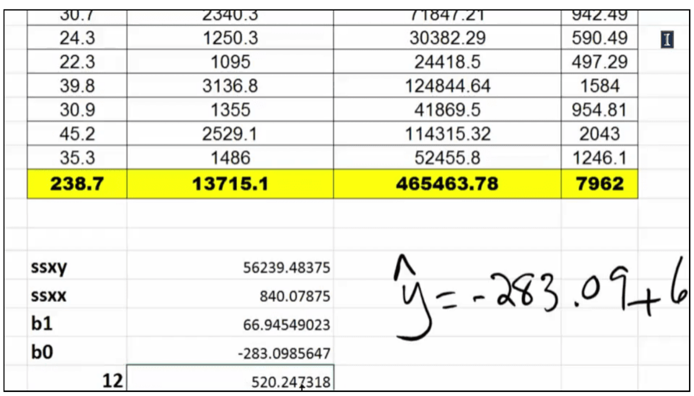
ssxy	56239.48375
ssxx	840.07875
b1	66.94549023
b0	-283.0985647
12	

The slide shows the outcome of b0, which is −283.0985647. In addition, the following formula is shown: $\hat{y} = -283.09 + 66.945x$.

Transcript

So, if I now use this value, our equation is ŷ is negative 283.09 plus 66.945 x. If I say my x-value is 12, then my regression equation would return the value of this intercept plus the value that I want to predict times b1, the slope, and it's 520.

Population Sales Excel Worksheet (8 of 8) - Slide 42



30.7	2340.3	71847.21	942.49
24.3	1250.3	30382.29	590.49
22.3	1095	24418.5	497.29
39.8	3136.8	124844.64	1584
30.9	1355	41869.5	954.81
45.2	2529.1	114315.32	2043
35.3	1486	52455.8	1246.1
238.7	**13715.1**	**465463.78**	**7962**

ssxy	56239.48375
ssxx	840.07875
b1	66.94549023
b0	-283.0985647
12	520.247318

The slide shows a new cell called 12 placed below the b0 cell. Here, this value is calculated by using the formula presented in Slide 41 Population Sales Excel Worksheet (7 of 8) and replacing x by 12. The value is 520.247318.

Download the Square Method Excel file (Refer to Least Square Method Calculation - Second worksheet)

Transcript

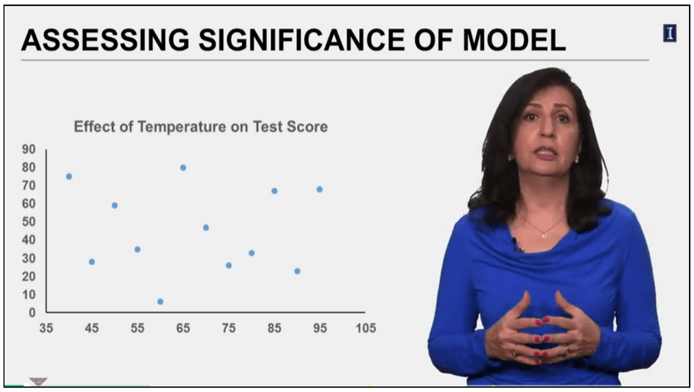
So, if I open up a store somewhere that has 12,000 people living nearby, I expect to sell $520,000 worth of annual sales.

Lesson 3-3 Assessing the Regression Model

Lesson 3-3.1 Assessing the Regression Model

[Media Player for Video](#)

Assessing Significance of Model - Slide 43

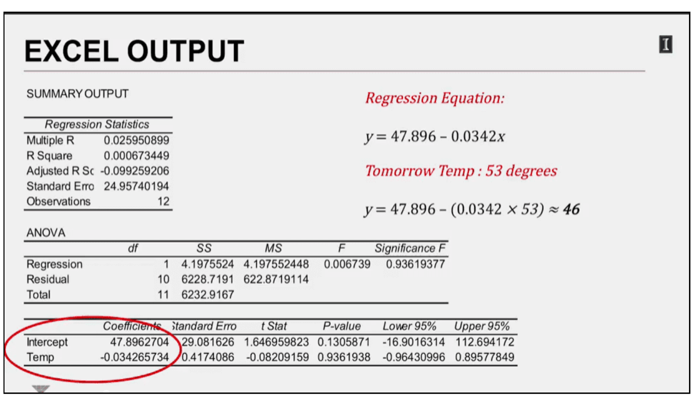


Scatter plot entitled, Effect of temperature on test score. The x-axis is from 35 to 105 and the y-axis from 0 to 90. The data points are randomly distributed on the plot, showing no correlation between the data.

Transcript

I'm going to start this lesson with an example that you know what I'm saying doesn't make sense. I'm doing it to make a point, so stay with me. Let's say, a student thinks that her test score is influenced by the temperature outside. For this, she collected date, some data and sense she has just learned about regression, she develops an equation and now she's happy that she can predict her test scores beforehand. Like I said, you know that this will not be the case and here is the scatter plot of the data she has. This is not looking like a pattern. However, if she used Excel to run the regression, she would get the following.

Excel Output - Slide 44



Summary Output

Regression Statistics - Slide 44

Multiple R	0.025950899
R Square	0.000673449
Adjusted R Square	−0.099259206
Standard Error	24.95740194
Observations	12

ANOVA - Slide 44

	df	SS	MS	F	Significance F
Regression	1	4.1975524	4.197552448	0.006739	0.93619377
Residual	10	6228.7191	622.8719114		
Total	11	6232.9167			

Other statistics - Slide 44

	Coefficients	Standard error	t-stat	p-value	Lower 95%	Upper 95%
Intercept	**47.8962704**	29.081626	1.646959823	0.1305871	−16.9016314	112.694172
Temp	**−0.034265734**	0.4174086	−0.08209159	0.9361938	−0.96430996	0.89577849

The coefficients of Intercept and Temp are highlighted.

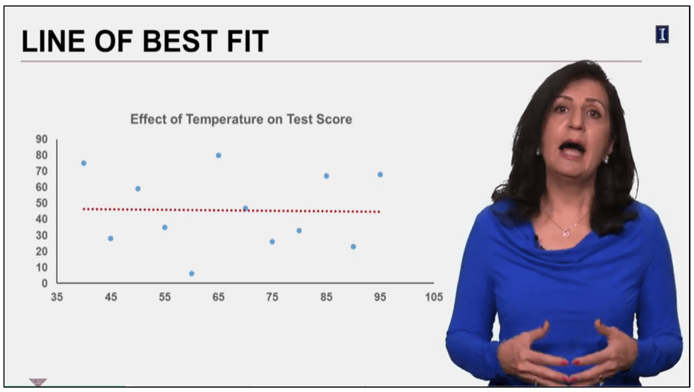
Regression Equation: $y = 47.896 - 0.0342x$

Tomorrow Temp: 53 degrees $y = 47.896 - (0.0342 \times 53) \approx 46$

Transcript

Focusing, on the part of the table that gives the slope and the intercept, then the regression equation is 47.896 − 0.0342 times x where x is the temperature outside. The weather forecast says, tomorrow the high will be 53 degrees, so the student can expect the test score of 46. How could I convince her that she has a bad model at hand and this should not be you used? This is the point I want you to get in this lesson. In real life, there are instances that the analysis is worthless, but they're just not as easy to detect as my overly simple and ridiculous example here. First, you should know that no matter what the data you use in your regression analysis, Excel will always find the line that fits those observations using the least square method. In this case, the line will be this, but just because we have an equation for line, doesn't mean that we have found something meaningful.

Line of Best Fit - Slide 45

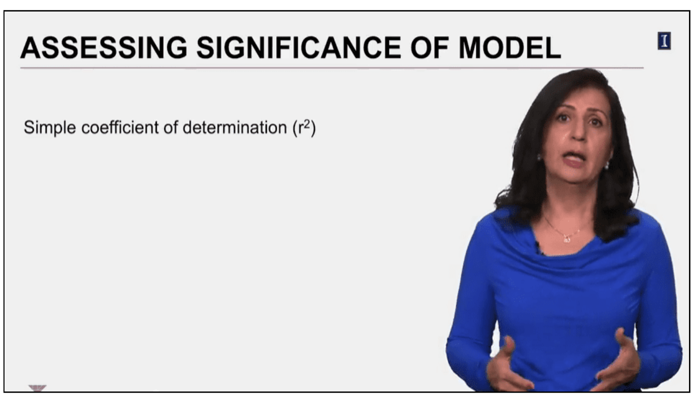


The slide shows the same scatter plot as Slide 43 Assessing Significance of Model. Here, a flat dotted regression line is added at around y = 47.89.

Transcript

So, in general, how do we detect a model which is poor versus a model which is good? We now turn to the question of how useful is a particular regression model?

Assessing Significance of Model - Slide 46

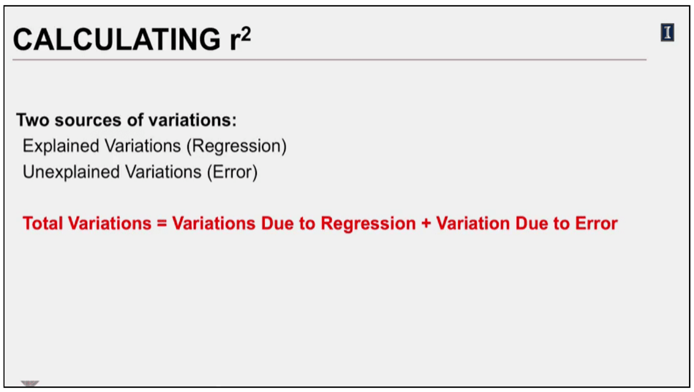


Simple coefficient of determination r^2

Transcript

After all, a model that explains 90% of the variation is more useful than one that explains only 10% of the variations. One measure of usefulness of a regression model is the simple coefficient of determination. This coefficient is represented by R-squared. And most of the time, it is just called R-squared and not by its proper name of simple coefficient of determination. It is important to know how we calculate R-squared, so that it's meaning will become clear. The sources of variation when performing regression are from two sources, one is from the regression variable we have identified. We call this the explain variation and the other source which is unexplained are called error or residuals and this is everything else that causes variation.

Calculating r^2 (1 of 2) - Slide 47



Two sources of variation:

Explained Variations (Regression)

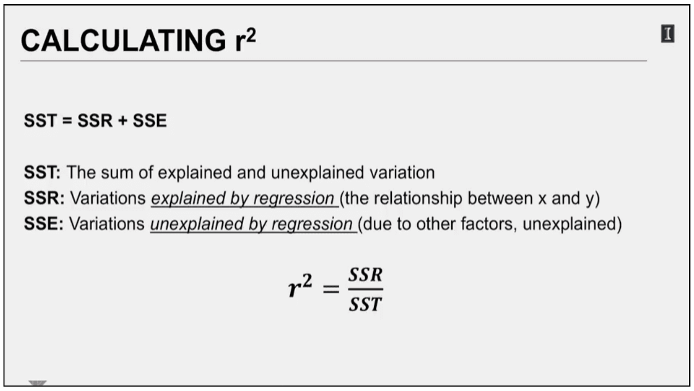
Unexplained Variations (Error)

Total variations = variations due to regression + variations due to error

Transcript

So, think about it this way, total variation is the sum of these two sources. If we have identified the variable which is influencing the variable with a strong predicted ability, the majority of the variations we see in the response variable should be due to the regression variable. But if we have identified a variable which has very little influence on the response variable, then the majority of the variation we observe in the response variable will be due to other sources, what we call unexplained sources. The notation for total variation is this, first you see SS in all parts. Again, SS stands for sum of squares. This is the formula for total variation. SST, which stands for sum of square of total is the sum of the squared differences between observed y-value and the average y-value.

Calculating r^2 (2 of 2) - Slide 48



SST = SSR + SSE

strong>SST: The sum of explained and unexplained variation

SSR: Variations *explained by regression* (the relationship between x and y)

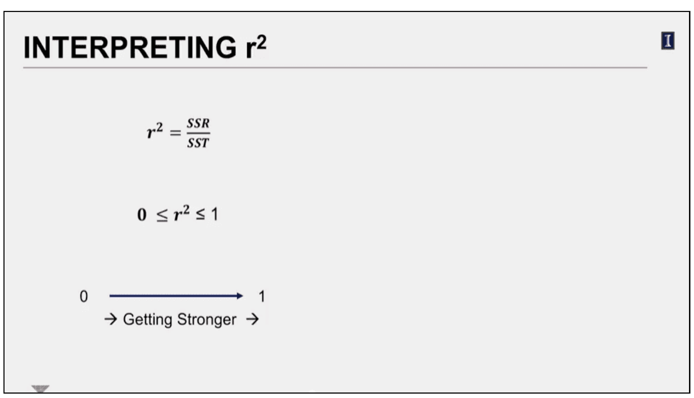
SSE: Variations *unexplained by regression* (due to other factors, unexplained)

$r^2 = \frac{SSR}{SST}$

Transcript

SSR, which is sum of square due to regression variable shows the variation that is explained by the regression model. And SSE, is the sum of square due to error which is the variation that is unexplained by the regression model we have, R-squared, the simple coefficient of determination, is the ratio of the explained variations to the total variation, R-squared being the ratio, will always be a value between zero and 1.

Interpreting r^2 - Slide 49



$r^2 = \frac{SSR}{SST}$

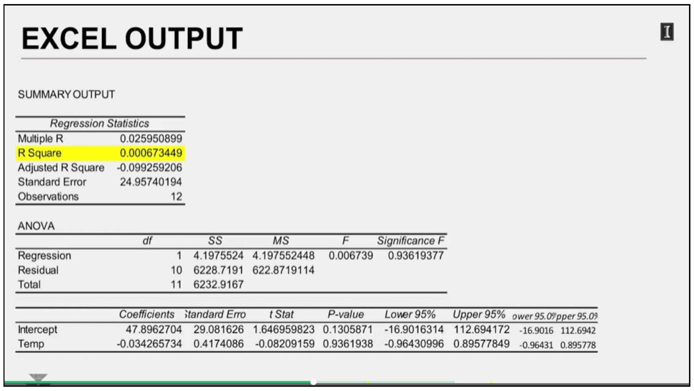
$0 \leq r^2 \leq 1$

There is an arrow pointing from 0 to 1 which means getting stronger.

Transcript

The closer R-squared is to 1, the stronger is the regression model. A regression model with an R-squared of 90% means that 90% of the variations in the response variable, y can be explained by the independent variable x.

Excel Output - Slide 50



EXCEL OUTPUT

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.025950899
R Square	0.000673449
Adjusted R Square	-0.099259206
Standard Error	24.95740194
Observations	12

ANOVA

	df	SS	MS	F	Significance F
Regression	1	4.1975524	4.197552448	0.006739	0.93619377
Residual	10	6228.7191	622.8719114		
Total	11	6232.9167			

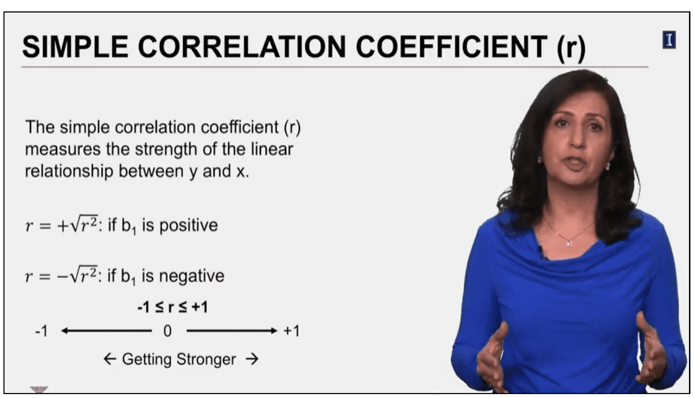
	Coefficients	Standard Erro	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	47.8962704	29.081626	1.646959823	0.1305871	-16.9016314	112.694172	-16.9016	112.6942
Temp	-0.034265734	0.4174086	-0.08209159	0.9361938	-0.96430996	0.89577849	-0.96431	0.895778

The slide shows the same tables as Slide 44 Excel Output. Here, the R square row in the Regression statistics table is highlighted.

Transcript

Let's look back at the Excel output from our example of effective temperature on test score. Focus on the top table, the highlighted field that is R-squared for our example. It is 0.0006, extremely small and close to zero. This is a good reason to throw this model away. Now, let's practice. This is the output for the example we used in the previous lesson. Effective population and annual sales. What is your opinion of this regression model? How much of the annual sale differences between stores is explained by the size of the population living near the stores and how much by other things? R-squared here is 0.7131. So, about 71% of annual sales can be explained by the population living near that stores location, so this is a very good model. However, there are other sources of variations and collectively, they explain about 29% of the variations. You often hear and may even use the word correlation. In linear regression, simple correlation coefficient measures the strength of the linear relationship between y and x and is denoted by r and is calculated by taking the square root of R-squared.

Simple Correlation Coefficient (r) - Slide 51



The simple correlation coefficient (r) measures the strength of the linear relationship between y and x.

$r = +\sqrt{r^2}$: if b_1 is positive

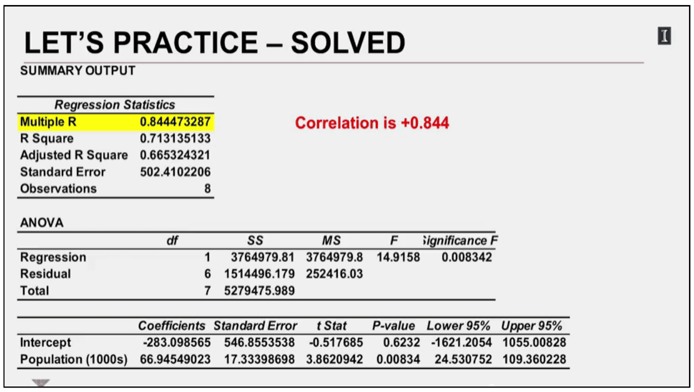
$r = -\sqrt{r^2}$: if b_1 is negative

There is a scale from −1 to 1. The scale has two arrows that represent "Getting stronger." The arrow to the right is pointing from 0 to 1 and the arrow to the left is pointing from 0 to −1. Above the scale is the formula: $-1 \leq r \leq 1$

Transcript

Correlation which is denoted by r, is positive if slope is positive which means x has a positive coefficient and we say x and y are positively correlated. If the coefficient of x is negative, then we say they are negatively correlated. Correlation can take on any value between negative 1 and 1. The closer to 1, the stronger the relationship.

Let's Practice - Solved - Slide 52



Regression Statistics	
Multiple R	0.844473287
R Square	0.713135133
Adjusted R Square	0.665324321
Standard Error	502.4102206
Observations	8

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3764979.81	3764979.8	14.9158	0.008342
Residual	6	1514496.179	252416.03		
Total	7	5279475.989			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-283.098565	546.8553538	-0.517685	0.6232	-1621.2054	1055.00828
Population (1000s)	66.94549023	17.33398698	3.8620942	0.00834	24.530752	109.360228

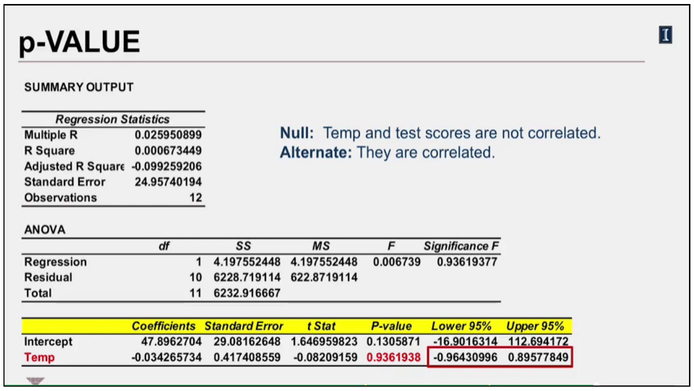
The slide shows the same tables as [Slide 32 Excel Output (1 of 3)](). Here, the Multiple R row in the Regression statistics table is highlighted. In addition, it says: **Correlation is +0.844.**

Transcript

In Excel, multiple r shows the correlation and it appears in the top table. In this example, where we land an analysis on effective temperature on test score, not surprisingly r is very close to zero. It's about 0.026 which is by the way, the square root of 0.0006 which is our R-squared and the sign comes from the coefficient of the temperature. In this case it's negative. The output of regression model for effective population and annual sales is shown here. What is the direction and magnitude of correlation between these two variables? The two variables are positively correlated because the coefficient for population is positive and its magnitude is 0.844, suggesting a fairly strong positive correlation, between population and annual sales at a store. So, we started this lesson with an obvious example of taking two variables that had no relationship, so I can explain how can assess if we have a good model or not.

So, now you know if R-squared is small and the variables are not correlated, then the model is useless. But we can find the answer to this question in other parts of the output also. Look closely at everything that is displayed in the third table, especially the values for the independent variable, temp. When you run regression analysis, you're building on the topic of hypothesis testing. Regression always assumes, that the independent variables you have identified have no relationship. So, if you get a small p-value less than the alpha of significance then you will be rejecting the null hypothesis. In the case of temperature and the test score, the p-value for temperature is quite large, 0.9361, so we will not reject the hypothesis, the temperature and test scores are not correlated.

p-value - Slide 53



The slide shows the same tables as the ones presented in Slide 44 Excel Output. Here, the first row of the Other statistics table is highlighted, the Temp p-value is colored red, and the Temp Lower 95% and Upper 95% are circled.

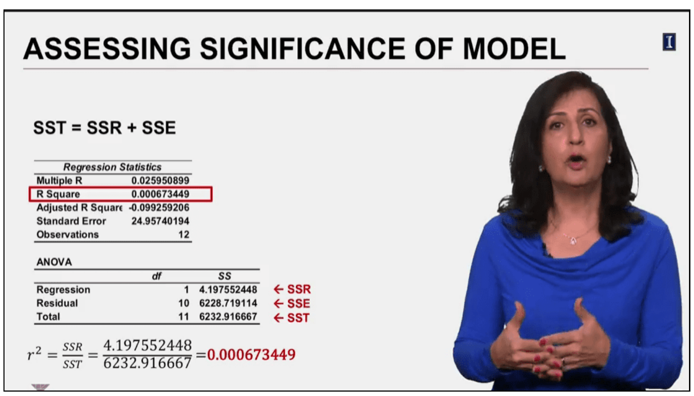
Null: Temp and test scores are not correlated

Alternate: They are correlated.

Transcript

You also see that Excel is providing the confidence interval, 95% in this case. For the coefficient of the temperature, focus on that role. The interval is between negative 0.964 to positive 0.895, zero is a value that is within the interval, so that could really be the coefficient for temperature. If you put zero as the coefficient of temperature, you would eliminate this variable all together from the equation.

Assessing Significance of Model - Slide 54



The slide shows the Regression Statistics and ANOVA tables as in Slide 44 Excel Output. The ANOVA table has just the df and SS columns. Here, in the Regression statistics table, the R square row is highlighted. In the ANOVA table, the regression SS value (4.1975524) is the SSR, the residual SS value (6228.7191) is the SSE, and the total SS value (6232.9167) is the SST.

$$r^2 = \frac{SSR}{SST} = \frac{4.1975524}{6232.9167} = 0.000673449$$

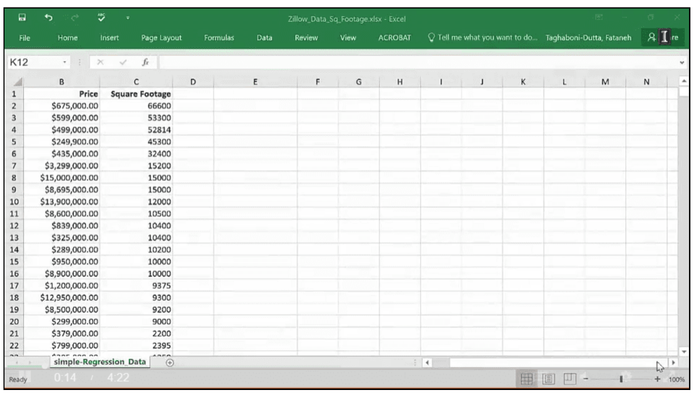
Transcript

So, now you see that we can tell that this was a bad model through many different measures we have here. Last part of the output which I have not mentioned is the table in the middle for ANOVA. ANOVA, stands for analysis of variance and is not part of the scope of this class, so I'm not going to be using this part of the output. However, I just want you to focus on one part of it and then we will move on. We learned that the total variations is the sum of these two sources, regression and other sources we collectively call error. Here I have taken small part of the Excel output to show you something from ANOVA part of it. Focus on the SS column that these values are for these notations. Using the formula for calculating R-squared and then taking the ration of SSR to SSE, you get the same value that you see in the very first table for the R-squared. Now that you understand how to first build the model and then assess the significance of the model, we can move on to practicing this and learning how to build effective, predictive models.

Lesson 3-3.2 Testing the Significance in Excel

Media Player for Video

Zillow Data Excel Worksheet (1 of 5) - Slide 55



	B	C
1	Price	Square Footage
2	$675,000.00	66600
3	$599,000.00	53300
4	$499,000.00	52814
5	$249,900.00	45300
6	$435,000.00	32400
7	$3,299,000.00	15200
8	$15,000,000.00	15000
9	$8,695,000.00	15000
10	$13,900,000.00	12000
11	$8,600,000.00	10500
12	$839,000.00	10400
13	$325,000.00	10400
14	$289,000.00	10200
15	$950,000.00	10000
16	$8,900,000.00	10000
17	$1,200,000.00	9375
18	$12,950,000.00	9300
19	$8,500,000.00	9200
20	$299,000.00	9000
21	$379,000.00	2200
22	$799,000.00	2395

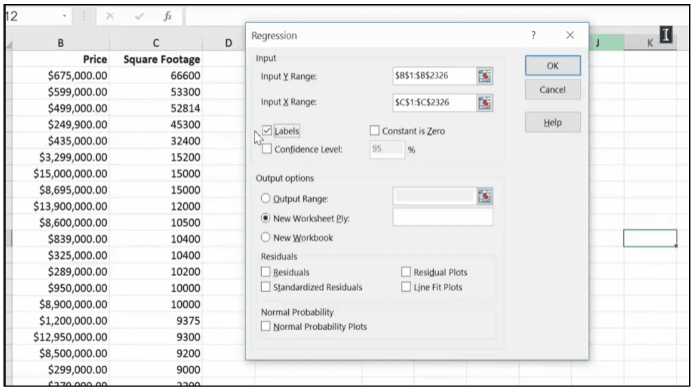
The slide shows the Zillow data set. The data includes two columns: Price and Square Footage.

Download the Zillow Data Square Footage Excel file (Refer to Data - First worksheet)

Transcript

So, here's another example that I'm going to use to show you simple linear regression using data analysis. This data was actually downloaded from Zillow for homes on sale in Chicago, and there are over 2,000 data elements in this file. Now, one of the things that we expect is that the square footage of a home impacts the price of the house. So, that the larger the house is, the higher the price of the home will be. So, I'm going to use a simple linear regression to check this, I'm going to go to data, I'm going to go to data analysis, and I'm going to pick regression. So, first of all, what is our y? Y is our price here. We are expecting price to be dependent on square footage. So, I'm going to pick price, and I'm going to control shift down and pick all of it. And as you can see, there are over 2,300 data points here. That's my y-value. Then I'm going to go up and pick the x-value, and that's the square footage of the home.

Zillow Data Excel Worksheet (2 of 5) - Slide 56

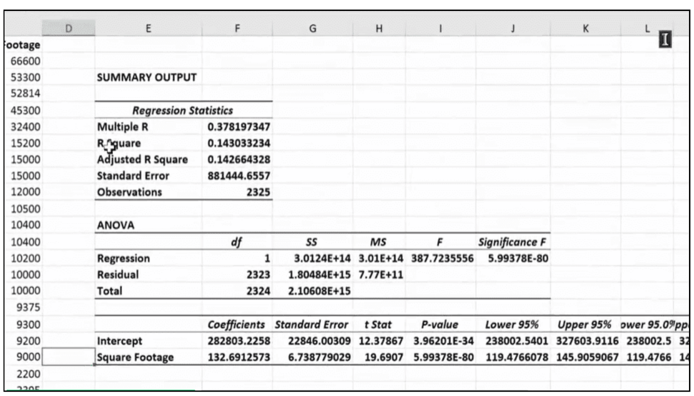


The slide shows what to include in the Regression pop-up window. Once going to the Data tab in the main menu, selecting the Data Analysis, and choosing the Regression option, another pop-up window appears. The window consists of four sections: Input, Output, Residuals, and Normal probability. In the Input section, the Input y range is $B$1:$B$2326, the Input x range is $C$1:$C$2326, the Labels checkbox option is selected, and the Confidence level is set to be 95%. In the Output section, the New worksheet ply radio option is selected.

Transcript

I'm going to say that I have labels, and this time, I'm going to just put the outputs right next to it. So, I'm going to click here, I'm going to click somewhere here, and say, "Okay". I'm going to expand my output, so here's our output.

Zillow Data Excel Worksheet (3 of 5) - Slide 57

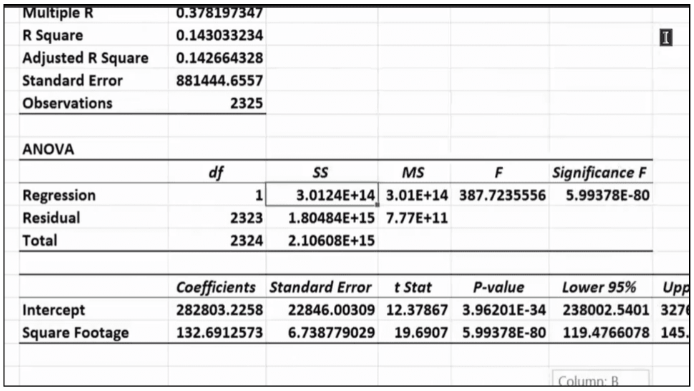


The slide shows the summary output distributed into three tables with their respective values. The first table is the regression statistics, the second table is the ANOVA, and the third table are other statistics that compare the Intercept with the Square footage.

Transcript

First of all, look at R-squared, in this case, 14.3. So, only 14.3% of the price of the home seems to be about how large it is. So, it doesn't seem to be a very good explanatory variable. Exactly what I had said before, if you look at here, the majority of it is coming from the residuals. Residuals are all the other sources, so if you think about the total variations that we see, which is shown here, only a small percentage of it is coming from regression, and the rest of it is coming from residuals.

Zillow Data Excel Worksheet (4 of 5) - Slide 58



Multiple R	0.378197347					
R Square	0.143033234					
Adjusted R Square	0.142664328					
Standard Error	881444.6557					
Observations	2325					

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3.0124E+14	3.01E+14	387.7235556	5.99378E-80
Residual	2323	1.80484E+15	7.77E+11		
Total	2324	2.10608E+15			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upp
Intercept	282803.2258	22846.00309	12.37867	3.96201E-34	238002.5401	327(
Square Footage	132.6912573	6.738779029	19.6907	5.99378E-80	119.4766078	145.

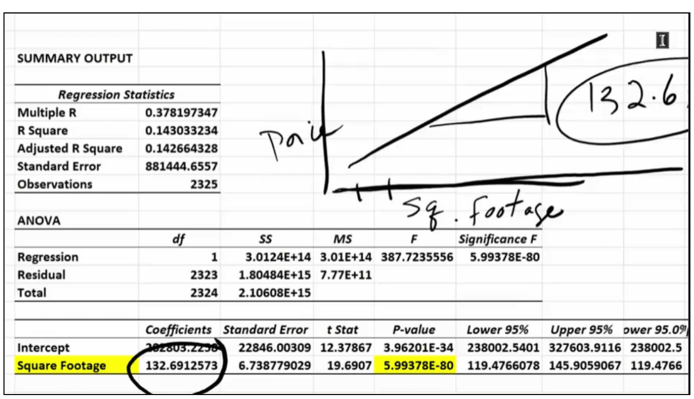
The slide shows the Regression SS value (3.0124E+14) in the ANOVA table selected.

Transcript

Remember, these are to the power of 14 and to the power of 15, so, there is a magnitude of difference between residuals and the SSR. However, if you look at what we see here, if you look at the p-value for the square footage, just focus on the p-value. The p-value for the square footage is extremely small. So, what it's saying to you is the following, that you have identified a variable, square footage, which has a significant relationship with the price of the home. However, on its own, it's not enough to explain the variabilities that you see in the prices of the home from one to the other. This is actually an example that I would expand on, when we would do multiple regression, so, this is a process.

Sometimes when we do a simple linear regression, we may find that we don't have a very good model, and that's when we have more than one thing influencing our dependent variable, the one that we want to predict. So, it doesn't mean that you would throw away all your analysis, because what I'm seeing is that I have come up with a good variable, a variable that has a very strong influence on what I'm trying to predict, price of the home, but on its own it's not enough. Now, what are you seeing here? That right now, based on what you're seeing, for every increase in the square footage, the price of the home will go up by $132. That's basically what it means, this is the rate of change. Remember, if I was drawing this, and it's a positive value, because it has no sign, and its correlation is about 37%. Remember, R, which is the correlation, can go between 0 and 1, to negative 1, so, the closer you are to 1, the higher is the correlation. So, they are correlated, they're not as strongly as we'd want it to be, and that shows in the R-squared, right? It's there are other things that are influencing the home's price. It could be its location, it could be the age of the home, how many bedrooms it has, how many bathrooms we have.

Zillow Data Excel Worksheet (5 of 5) - Slide 59



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.378197347
R Square	0.143033234
Adjusted R Square	0.142664328
Standard Error	881444.6557
Observations	2325

ANOVA

	df	SS	MS	F	Significance F
Regression	1	3.0124E+14	3.01E+14	387.7235556	5.99378E-80
Residual	2323	1.80484E+15	7.77E+11		
Total	2324	2.10608E+15			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%
Intercept	282803.2258	22846.00309	12.37867	3.96201E-34	238002.5401	327603.9116	238002.5
Square Footage	132.6912573	6.738779029	19.6907	5.99378E-80	119.4766078	145.9059067	119.4766

The slide shows the Square footage p-value (5.99378E-80) and coefficient (132.6912573) in the other statistics table highlighted. In addition, there is a graph where the x-axis is the Square footage and the y-axis is the price. The plot shows a slope of 132.69 with a positive regression.

[Download the Zillow Data Square Footage Excel file (Refer to Simple Regression - Second worksheet)](#)

Transcript

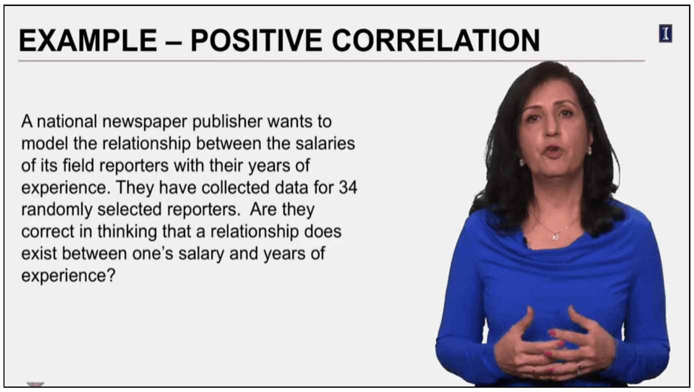
So, we can develop this further, but let me just focus on the coefficient of 132. So, what this is saying to you is that if this is the square footage, and this is the price of the home, the line is somewhere like this. The slope is 132.69, so, as you go by 1 point up, the price of the home will go up by $132.69. Now, since R-squared is so small, I am not going to use this output for a prediction. I'm not going to say, "Okay, now, if I say the house is 2,000 square feet, can you predict what the price of the home will be?" Because it's not a good predictive model, we just have identified a variable that has a very small p-value, which means it has a significant relationship with the price of the home.

Lesson 3-4 Using the Model

Lesson 3-4.1 Using the Model

Media Player for Video

Example: Positive Correlation (1 of 2) - Slide 60

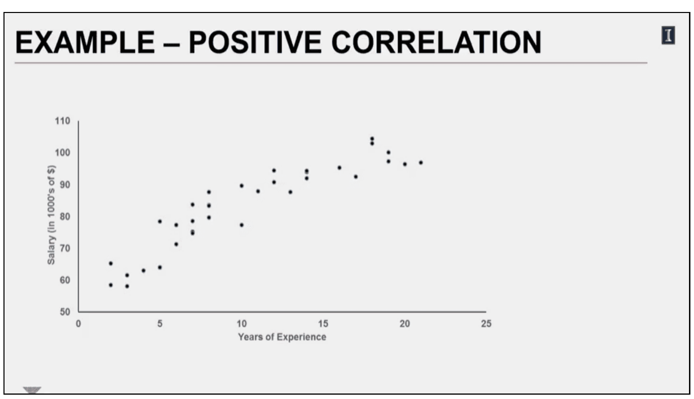


A national newspaper publisher wants to model the relationship between the salaries of its field reporters with their years of experience. They have collected data for 34 randomly selected reporters. Are they correct in thinking that a relationship does exist between one's salary and years of experience?

Transcript

A national newspaper publisher wants to model the relationship between the salaries of its field reporters with their years of experience. They have collected data for 34 randomly selected reporters. Are they correct in thinking that a relationship does exist between one's salary and years of experience? So here, the independent variable is the years of experience, that would-be x. And the response variable is the salary, and this would-be y. We will first begin by doing a scatter plot to see if we can detect a linear relationship between the two variables.

Example: Positive Correlation (2 of 2) - Slide 61



A scatter plot with the x-axis as the years of experience from 0 to 25 and the y-axis as the salary in $1000's of $ from 50 to 110. The plot shows that with the increase of experience, the salary increases corresponding, which forms a positive correlation.

Transcript

From the scatter plot, we get the sense for a strong positive correlation between the years of experience and one's salary. So, we can go ahead and run the data to the regression analysis. But before doing that, let me alert you to a common mistake, mixing up the two variables. This scatter plot shows a positive correlation too, but there's something major wrong with this. Can you detect it? If not, pay close attention to what variable is on the x-axis and what is on the y-axis. Salary, the dependent variable is being displayed on the x-axis, that's a no-no.

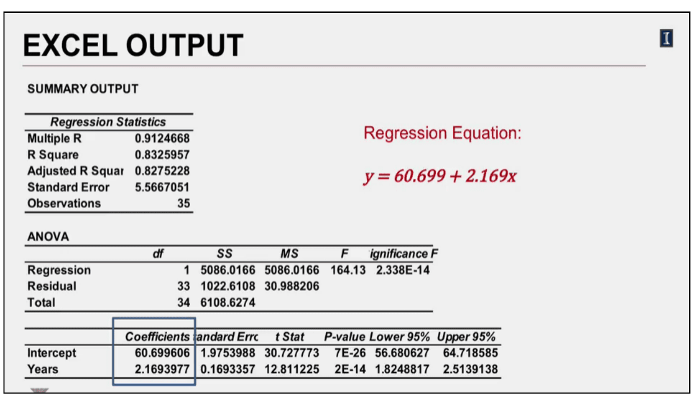
What is Wrong with This Picture? - Slide 62

A scatter plot with the x-axis as the salary in $1000’s of $ from 50 to 110 and the y-axis as the years of experience from 0 to 25. The plot shows that with the increase of salary, the years of experience increase, which correspond to a positive correlation.

Transcript

Just like I have said before, Excel is automating tasks for you. It will not be checking your model, no software will. You will have to be sure that you model the problem correctly. Otherwise, you will be looking at an invalid analysis. Okay, now we can go ahead and do it correctly and run the analysis.

Excel Output - Slide 63



Regression Statistics	
Multiple R	0.9124668
R Square	0.8325957
Adjusted R Squar	0.8275228
Standard Error	5.5667051
Observations	35

ANOVA

	df	SS	MS	F	ignificance F
Regression	1	5086.0166	5086.0166	164.13	2.338E-14
Residual	33	1022.6108	30.988206		
Total	34	6108.6274			

	Coefficients	andard Errc	t Stat	P-value	Lower 95%	Upper 95%
Intercept	60.699606	1.9753988	30.727773	7E-26	56.680627	64.718585
Years	2.1693977	0.1693357	12.811225	2E-14	1.8248817	2.5139138

Summary Output

Regression Statistics - Slide 63

Multiple R	0.9124668
R Square	0.8325957
Adjusted R Square	0.8275228
Standard Error	5.5667051
Observations	35

ANOVA - Slide 63

	df	SS	MS	F	Significance F
Regression	1	5086.0166	5086.0166	164.13	2.338E-14
Residual	33	1022.6108	30.988206		
Total	34	6108.6274			

Other statistics - Slide 63

	Coefficients	Standard error	t-stat	p-value	Lower 95%	Upper 95%
Intercept	60.699606	1.9753988	30.727773	7E-26	56.680627	64.718585
Years	2.1693977	0.1693357	12.811225	2E-14	1.8248817	2.5139138

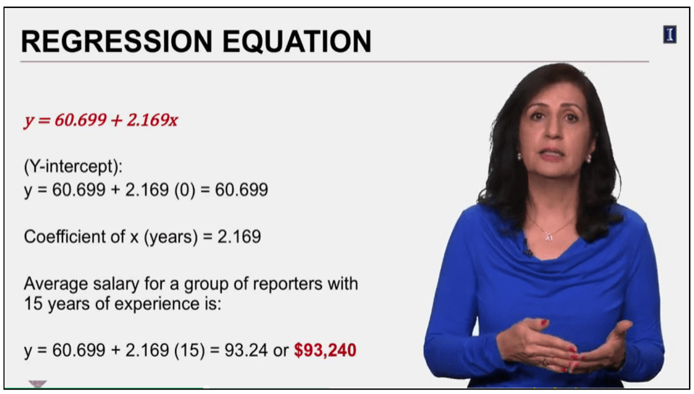
In the Other statistics table, the intercept and years coefficients are highlighted.

The regression equation is: $y = 60.699 + 2.169x$

Transcript

So, here's the output. First, let's see what percent of variations we see in someone's salary can we explain by their number of years of experience. First, let's see what percent of variations we see in someone's salary can we explain by their number of years of experience. The answer to this is in R-Squared. We can now see what the regression equation be, and that is found by looking here. So then, the equation is y is equal to 60.699 plus 2.169 times x. Now, let's understand this better. The intercept for this equation is 60.699. This means that if you put x equal to 0, this is the value y, then based on our data, this represents the predicted starting salary for a reporter in this organization. Then, we have the coefficient of independent variable, number of years of experience. This means for every year, the salary has gone up by 2.169, which is really $2,169, data was scaled by thousands. This 2.169 represents the average raise the reporters get per year.

Regression Equation - Slide 64



$y = 60.699 + 2.169x$

y intercept : $y = 60.699 + 2.169 \times (0) = 60.699$

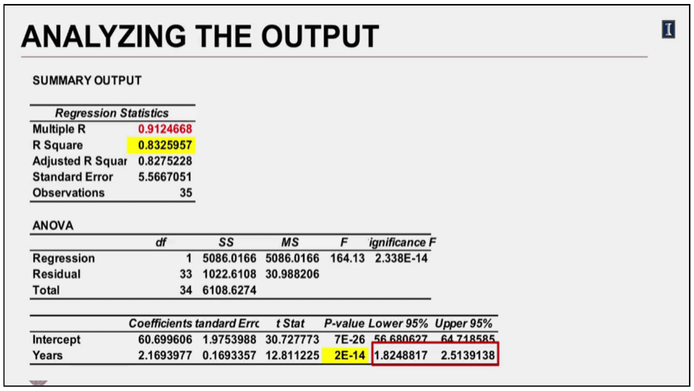
Coefficient of x (years) $= 2.169$

Average salary for a group of reporters with 15 years of experience is: $y = 60.699 + 2.169 \times (15) = 93.24$ or $93,240

Transcript

Now, if I ask you, what would you expect the average salary be, if I have a sample of reporters with an average of 15 years of experience, the answer will be 60.699 plus 2.169 times 15, which is 93.24. I will multiply this by 1,000 again because the salary data was entered in thousands of dollars and that would be $93,240.

Analyzing the Output - Slide 65



Regression Statistics	
Multiple R	0.9124668
R Square	0.8325957
Adjusted R Squar	0.8275228
Standard Error	5.5667051
Observations	35

ANOVA

	df	SS	MS	F	ignificance F
Regression	1	5086.0166	5086.0166	164.13	2.338E-14
Residual	33	1022.6108	30.988206		
Total	34	6108.6274			

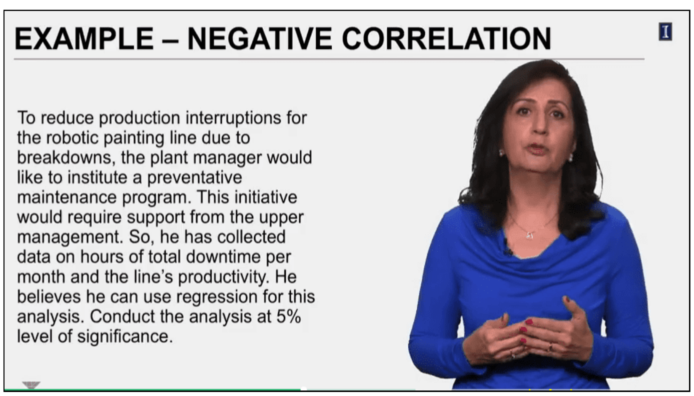
	Coefficients	tandard Errc	t Stat	P-value	Lower 95%	Upper 95%
Intercept	60.699606	1.9753988	30.727773	7E-26	56.680627	64.718585
Years	2.1693977	0.1693357	12.811225	2E-14	1.8248817	2.5139138

The slide shows the same tables as the ones presented in Slide 63 Excel Output. Here, in the Regression statistics table the Multiple R value (0.9124668) and the R square value (0.8325957) are highlighted. In the Other statistics table, the Years p-value (2E-14) and the Years Lower 95% value (1.8248817) and the Upper 95% value (2.5139138) are highlighted.

Transcript

Since we have such a high value of R-squared, it is not surprising that we get a very small p-value for years, highlighted in yellow. We have rejected the null hypothesis that one's years of experience has nothing to do with the person's salary. We also see in red font, that salary and years of experience are highly correlated at .91, this is a positive correlation. As we see the sign of the coefficient, for years, is positive. This means as the reporter's length of stay has increased, so has their salary. Again, on average, about $2,169 per year. Please be mindful, that the coefficient we have used is a point estimate, this is the mid-point, between the confidence interval for the coefficient, which means the two-average value of salary increase can be somewhere between $1,825 to $2,514. To do better, we should calculate the confidence interval for the prediction. This is not done very easily in Excel, though.

Example: Negative Correlation (1 of 2) - Slide 66

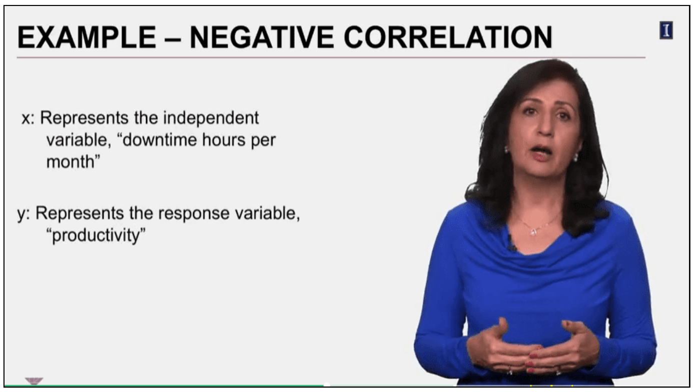


To reduce production interruptions for the robotic painting line due to breakdowns, the plant manager would like to institute a preventative maintenance program. This initiative would require support from the upper management. So, he has collected data on hours of total downtime per month and the line's productivity. He believes he can use regression for this analysis. Conduct the analysis at 5% level of significance.

Transcript

To reduce production interruptions for the robotic painting line due to breakdowns, the plant manager would like to institute a preventive maintenance program for the robots. This initiative will require support from the upper management. So, he has collected data on hours of total downtime per month and the line's productivity. He believes he can use regression for this analysis. In this study, the average number of hours the line is down, is the independent variable x, and the response variable is the productivity measure of the line. Scatter plot for the data is shown here.

Example: Negative Correlation (2 of 2) - Slide 67



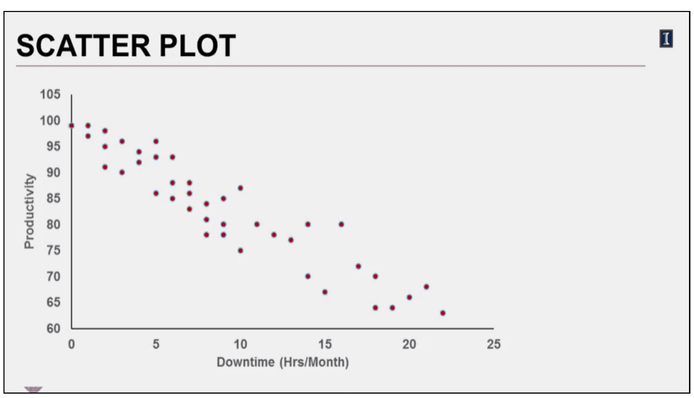
x: Represents the independent variable, "downtime hours per month"

y: Represents the response variable, "productivity"

Transcript

A strong negative correlation between the two variables as the number of hours the line is down increases, the productivity measure drops in a linear form.

Scatter Plot - Slide 68

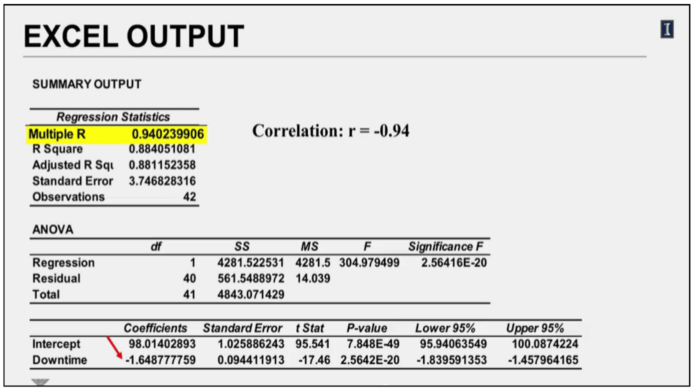


Scatter plot with the x-axis as the downtime hours per month from 0 to 25 and the y-axis as the productivity from 60 to 105. With the increase of downtime, the corresponding productivity decreases, which forms a strong negative correlation.

Transcript

So, we can apply simple linear regression to explore this relationship further.

Excel Output (1 of 2) - Slide 69



EXCEL OUTPUT

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.940239906
R Square	0.884051081
Adjusted R Squ	0.881152358
Standard Error	3.746828316
Observations	42

Correlation: r = -0.94

ANOVA

	df	SS	MS	F	Significance F
Regression	1	4281.522531	4281.5	304.979499	2.56416E-20
Residual	40	561.5488972	14.039		
Total	41	4843.071429			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	98.01402893	1.025886243	95.541	7.848E-49	95.94063549	100.0874224
Downtime	-1.648777759	0.094411913	-17.46	2.5642E-20	-1.839591353	-1.457964165

Summary Output

Regression Statistics - Slide 69

Multiple R	0.940239906
R Square	0.884051081
Adjusted R Square	0.881152358
Standard Error	3.746828316
Observations	42

ANOVA - Slide 69

	df	SS	MS	F	Significance F
Regression	1	4281.522531	4281.5	304.979499	2.56416E-20
Residual	40	561.5488972	14.039		
Total	41	4843.071429			

Other statistics - Slide 69

	Coefficients	Standard error	t-stat	p-value	Lower 95%	Upper 95%
Intercept	98.01402893	1.025886243	95.541	7.848E-49	95.94063549	100.0874224
Downtime	−1.648777759	0.094411913	−17.46	2.5642E-20	−1.839591353	−1.457964165

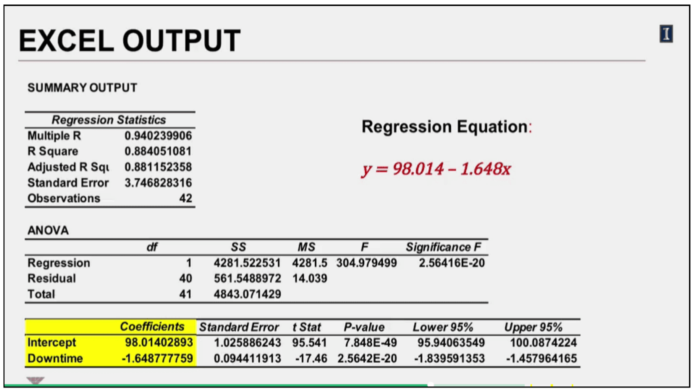
In the Regression statistics table, the Multiple R row is highlighted. In addition, there is an arrow pointing at the Downtime coefficient value (−1.648777759).

Correlation: r = −0.94

Transcript

We have used Excel to analyze data. The degree of correlation between average hours of down time in a month and productivity are negatively correlated. And R, the correlation is negative .94.

Excel Output (2 of 2) - Slide 70



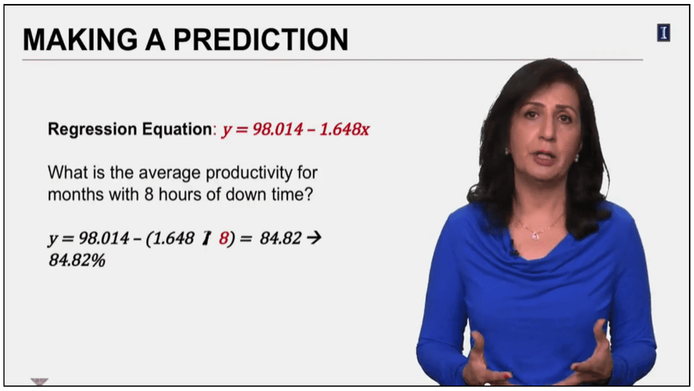
The slide shows the same tables as the ones presented in Slide 69 Excel Output. Here, the Intercept coefficient (98.01402893) and the Downtime coefficient (−1.648777759) are highlighted.

Regression equation: **y = 98.014 − 1.648x**

Transcript

Again, the sign comes from the sign for the downtime coefficient. R-squared shows that 88.4 percent of variations in productivity, we've seen our data, can explained by the regression variable downtime. The remaining 11.6 percent are due to other sources we call error. So, given that we have a strong relationship, we can go ahead and develop the regression equation. To develop the regression equation, we find the coefficients in the third part of the output. So, the equation is 98.0143 minus 1.648 times x. Looking at this equation, which is based on our data, the baseline for productivity is about 98%. And for every hour the line is down, the productivity goes down by an additional 1.648%. The manager believers that on average, the line is down for about 8 hours per month. If this is indeed the case, then what is the average productivity expected from the production line for the months with this much downtime? To find the predicted productivity we put value of 8 in variable x and that will give us the average productivity to be 84.82%.

Making a Prediction - Slide 71



Regression equation: **y = 98.014 − 1.648x**

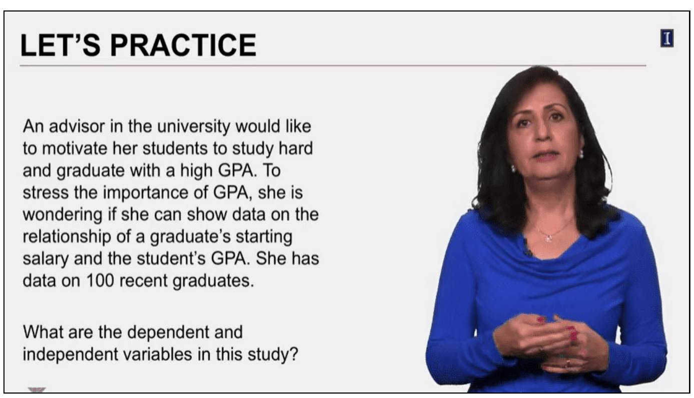
What is the average productivity for months with 8 hours of down time?

y = 98.014 − (1.648 × 8) = 84.82 → **84.82%**

Transcript

So, making the line more reliable, by instituting a preventative maintenance program, will reduce breakdowns. And that will improve productivity. Now, the plant manager can use this insight and do some economic analysis comparing the cost of maintenance program versus the cost associated with lost productivity. As you can see, the regression did not only have the ability to infer about the population, but now we can predict the value of the response variable for a given independent variable. Let's practice.

Let's Practice - Slide 72



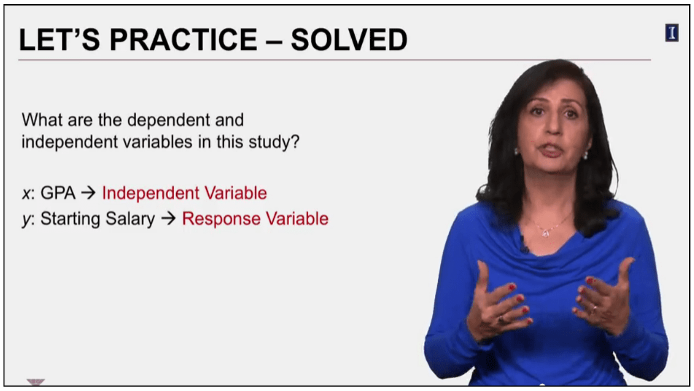
An advisor in the university would like to motivate her students to study hard and graduate with a high GPA. To stress the importance of GPA, she is wondering if she can show data on the relationship of a graduate's starting salary and the student's GPA. She has data on 100 recent graduates.

What are the dependent and independent variables in this study?

Transcript

An advisor in the university would like to motivate her students to study hard and graduate with a high GPA. To stress the importance of GPA, she is wondering if she can show data on the relationship of a graduate's starting salary and the students GPA. She has collected data on 100 recent graduates. What are the dependent and independent variables in this study?

Let's Practice - Solved - Slide 73



What are the dependent and independent variables in this study?

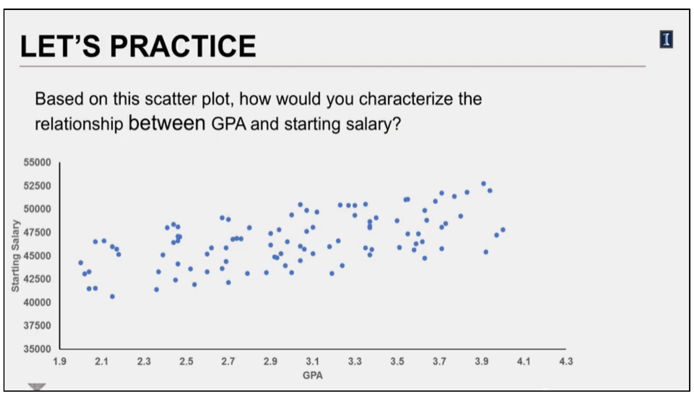
x: GPA → **Independent variable**

y: Starting salary → **Response variable**

Transcript

The independent variable is the student's GPA which the advisor believes influences starting salary, so that would be the y-variable, which is the response variable.

Let's Practice (1 of 2) - Slide 74



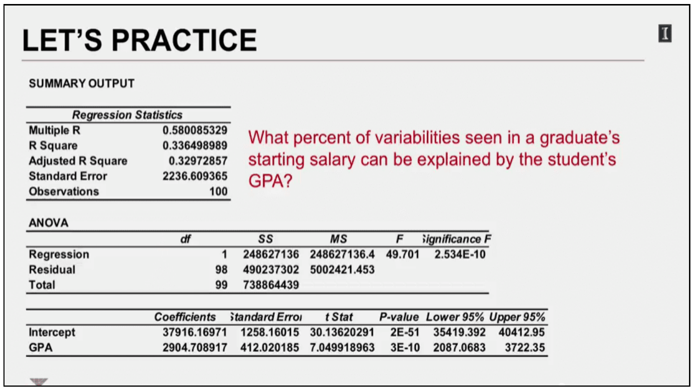
Based on this scatter plot, how would you characterize the relationship between GPA and starting salary?

The slide shows a scatter plot with the x-axis as GPA from 1.9 to 4.0 and the y-axis as the starting salary from 35000 to 55000. The plot shows that with the increase of GPA, the corresponding starting salary increases gently.

Transcript

Now, based on this scatter plot, how would you characterize the relationship with the GPA and starting salary?

Let's Practice (2 of 2) - Slide 75



Regression Statistics	
Multiple R	0.580085329
R Square	0.336498989
Adjusted R Square	0.32972857
Standard Error	2236.609365
Observations	100

ANOVA

	df	SS	MS	F	Significance F
Regression	1	248627136	248627136.4	49.701	2.534E-10
Residual	98	490237302	5002421.453		
Total	99	738864439			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	37916.16971	1258.16015	30.13620291	2E-51	35419.392	40412.95
GPA	2904.708917	412.020185	7.049918963	3E-10	2087.0683	3722.35

Summary Output

Regression Statistics - Slide 75

Multiple R	0.580085329
R Square	0.336498989
Adjusted R Square	0.32972857
Standard Error	2236.609365
Observations	100

ANOVA - Slide 75

	df	SS	MS	F	Significance F
Regression	1	248627136	248627136.4	49.701	2.534E-10
Residual	98	490237302	5002421.453		
Total	99	738864439			

Other statistics - Slide 75

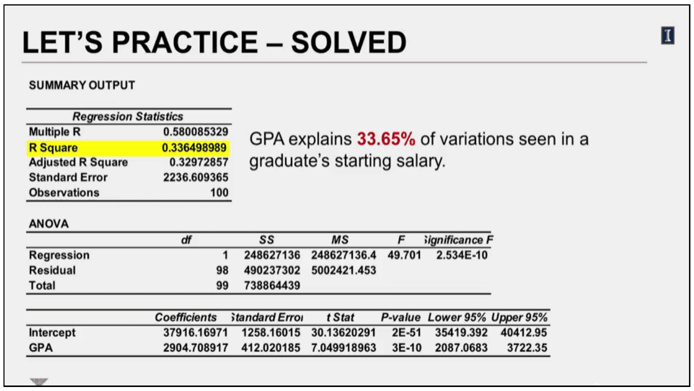
	Coefficients	Standard error	t-stat	p-value	Lower 95%	Upper 95%
Intercept	37916.16971	1258.16015	30.13620292	2E-51	35419.392	40412.95
GPA	2904.708917	412.020185	7.049918963	3E-10	2087.0683	3722.35

What percent of variabilities seen in a graduate's starting salary can be explained by the student's GPA?

Transcript

There appears to be a strong and positive relationship between GPA and starting salary. What percent of variability seen in a student's starting salary can be explained by the student's GPA? Based on our analysis, R-squared is about .3365, so GPA explains 33.65% of the variations we see in the starting salaries in our data, which means that remaining variation we see is due to other sources, unexplained by our model. So, is this advisor wrong?

Let's Practice - Solved - Slide 76



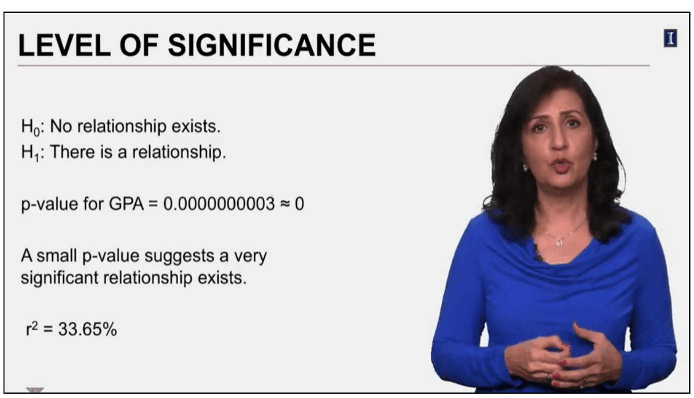
The slide shows the same tables as the ones presented in Slide 75 Let's Practice (2 of 2). Here, the R square row in the Regression statistics table is highlighted.

GPA explains **33.65%** of variations seen in a graduate's starting salary.

Transcript

Does GPA appear to be not that important? Actually, the answer to that is here in the p-value, and this value here is practically 0. Recall that the regression is running a hypothesis test on the independent variable. Now hypothesis is that no relationship exists. We can think of regression as a cynic. You have to have data to change its mind. In this case, GPA has such a small p-value that you will reject the null hypothesis at any level of significance.

Level of Significance - Slide 77



H_0:No relationship exists.

H_1:There is a relationship.

p-value for GPA: $0.0000000003 \approx 0$

A small p-value suggests a very significant relationship exists.

$r^2 = 33.65\%$

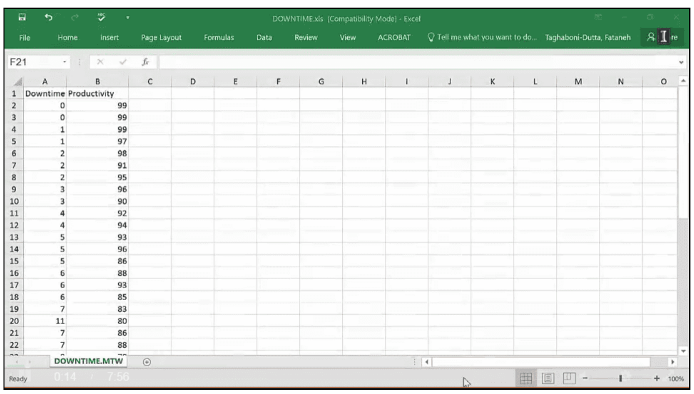
Transcript

So then why is R-squared not higher? What R-squared is saying is that while about a third of differences observed in a graduate's salary was explained by GPA, two thirds are due to other sources. Based on the p-value, you have identified one very significant source and that's good. But your regression equation is not very good because most of the day, ability is due to other factors. For example, could one of those factors be student's major? You would expect that to be a factor, wouldn't you? So, if the advisor needs to expand her data collection and include more independent variables, but then you will no longer be doing a simple linear regression, but a multiple regression. This is a topic of the last module in this class and I will come back to this later on.

Lesson 3-4.2 Simple Linear Regression: Excel Output Analysis

[Media Player for Video](#)

Downtime Excel Worksheet (1 of 11) - Slide 78



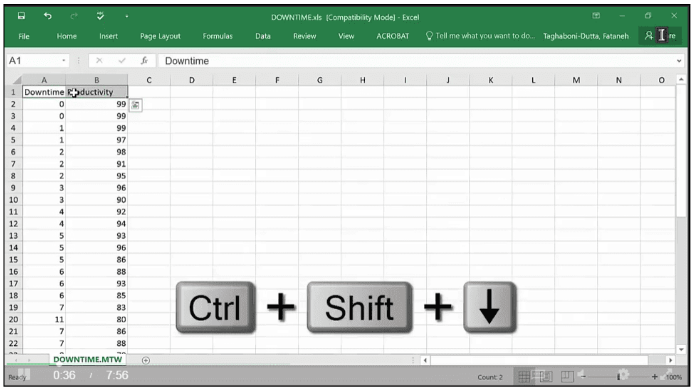
The slide shows the Downtime data set. The data includes two columns: Downtime and Productivity.

[Download the Downtime Excel file (Refer to Data - First worksheet)](#)

Transcript

In this example, I'm going to show the steps in order to run regression using Data Analysis. This is an example from our PowerPoint where the production line is believed to be influenced by the downtime. The productivity is influenced by how much the line goes down. So, if you have data for many days and we are going to analyze it. The first thing we want to do is to visually check whether or not this is going to be a linear regression or linear model that will fit.

Downtime Excel Worksheet (2 of 11) - Slide 79

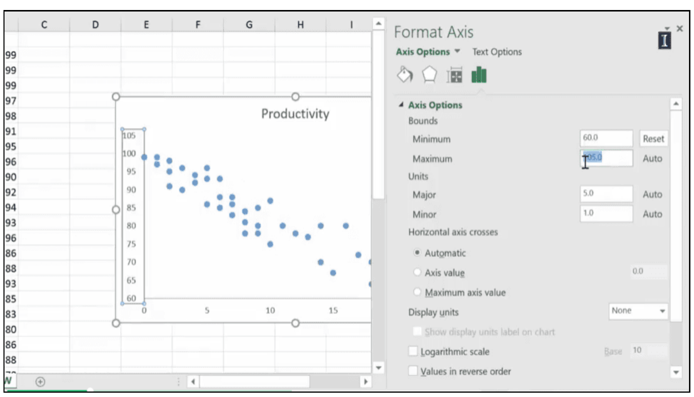


The slide shows how to select both columns of the data by going to the first row of these columns and pressing Ctrl + Shift + the down arrow.

Transcript

I'm going to pick my variables, go to Insert, Scatter Plot and plot these out. So, what I have done in the PowerPoint is that I formatted these so it would be a little easier to read. So, what you can see is that we don't have productivity down below, so one other thing I can do is change the access that it doesn't go from zero, so maybe it goes from 60 to something like 105.

Downtime Excel Worksheet (3 of 11) - Slide 80



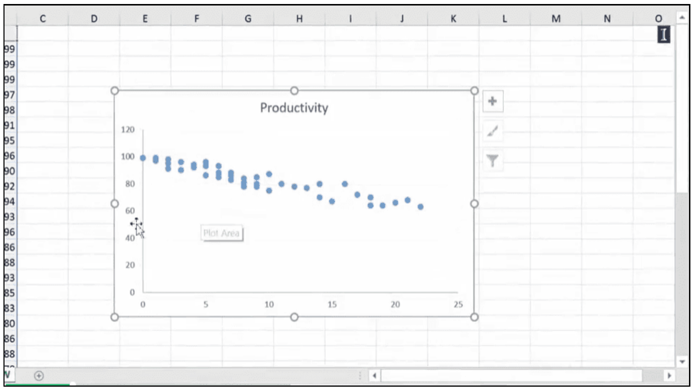
The slide shows the scatter plot created from selecting the Downtime and Productivity columns. The plot shows a strong negative correlation. After double clicking the plot, a right-hand panel called Format Axis appears. Inside this panel, the fourth axis option called Axis options is opened. This axis option has four sections: bounds, units, horizontal axis crosses, and display units. In the Bound section, the minimum option is 60 and the maximum option is 105. In the Units section, the major option is 5 and the minor option is 1. In the Horizontal axis crosses section, the automatic radio option is selected. Finally, in the Display units section, the unfolding option is set to be none.

Download the Downtime Excel file (Refer to Scatter plot - Second worksheet)

Transcript

And it automatically looks a little bit more spread out and you can see it a little bit better.

Downtime Excel Worksheet (4 of 11) - Slide 81

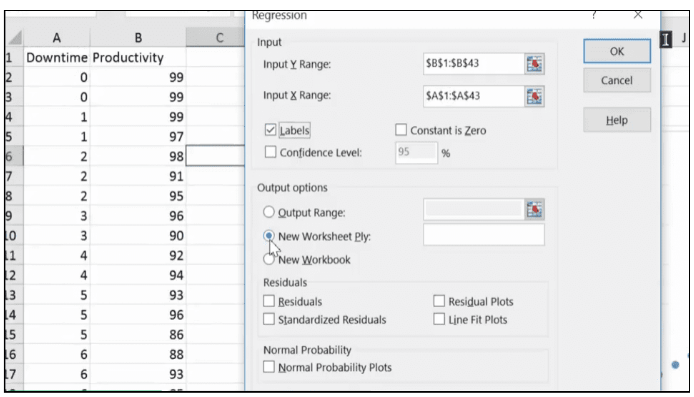


The slide shows a scatter plot with the x-axis as downtime from 0 to 25 and the y-axis as productivity from 0 to 120. Here, the data looks flatter than with the y range of 60 to 105.

Transcript

So, what you can is that it indeed looks more or less like a scattered plot that is linear in its form. Now, by the way, you can go ahead and have it inserted in the equation but I really don't think that that's the best way for you to come up with a linear equation and then analyze it because it doesn't give you all the elements we want. So, I really want to do this only for visual check. Now that I feel like I can apply linear regression, I'm going to go to Data, Data Analysis, when the window comes up, go down, pick Regression. Regression comes up. So, the first thing it asks for is the input of y-range, this is what you're trying to predict, so that's my productivity and I like to pick the labels also, so I'm going to click here, control shift, down, pick the entire data set, then go to input of x-range, so that's the downtime, control shift down again.

Downtime Excel Worksheet (5 of 11) - Slide 82

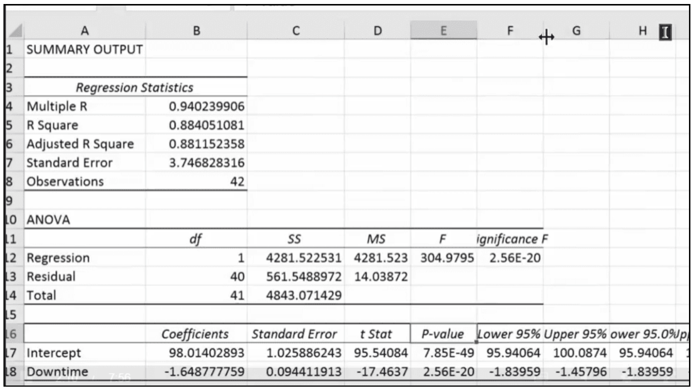


The slide shows what to include in the Regression pop-up window. Once going to the Data tab in the main menu, selecting Data Analysis, and choosing the Regression option, another pop-up window appears. The window consists of four sections: Input, Output, Residuals, and Normal probability. In the Input section, the Input y range is $B$1:$B$43, the Input x range is $A$1:$A$43, the Labels checkbox option is selected, and the Confidence level is set to be 95%. In the Output section, the New worksheet ply radio option is selected.

Transcript

I have already told it that it has labels and a new worksheet is what I wanted to have. So, here is my output. And as I mentioned, it automatically returns three different tables. So, I'm going to go over these with you.

Downtime Excel Worksheet (6 of 11) - Slide 83



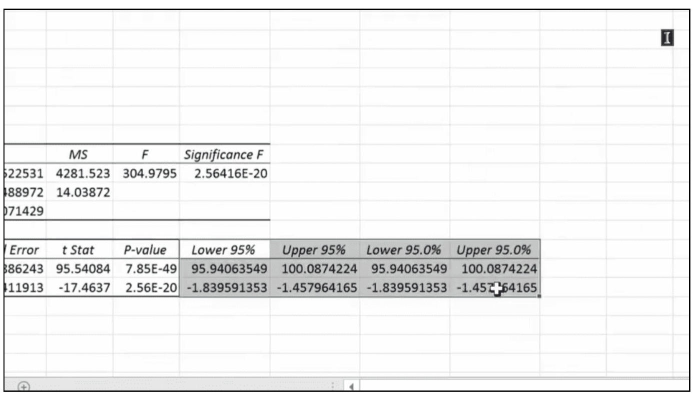
	A	B	C	D	E	F	G	H
1	SUMMARY OUTPUT							
2								
3	*Regression Statistics*							
4	Multiple R	0.940239906						
5	R Square	0.884051081						
6	Adjusted R Square	0.881152358						
7	Standard Error	3.746828316						
8	Observations	42						
9								
10	ANOVA							
11		*df*	*SS*	*MS*	*F*	*ignificance F*		
12	Regression	1	4281.522531	4281.523	304.9795	2.56E-20		
13	Residual	40	561.5488972	14.03872				
14	Total	41	4843.071429					
15								
16		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*
17	Intercept	98.01402893	1.025886243	95.54084	7.85E-49	95.94064	100.0874	95.94064
18	Downtime	-1.648777759	0.094411913	-17.4637	2.56E-20	-1.83959	-1.45796	-1.83959

The slide shows the summary output distributed into three tables with their respective values. The first table is the regression statistics, the second table is the ANOVA, and the third table are other statistics that compare the Intercept with the Downtime.

Transcript

Now, one of the things you can see is that the columns are not large enough to show all of this, so one of the best things we can do is just click here in between the columns and double click and it will expand to what you need.

Downtime Excel Worksheet (7 of 11) - Slide 84



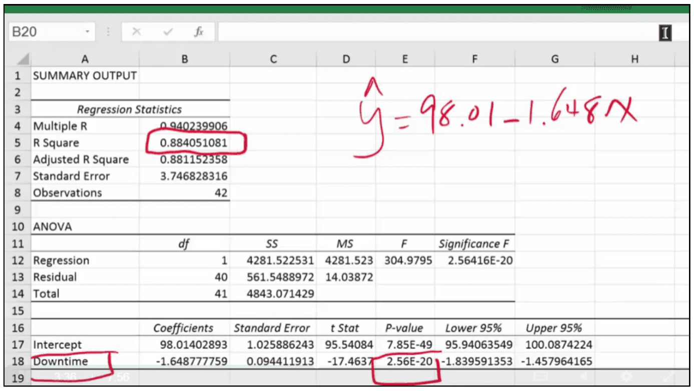
	MS	*F*	*Significance F*				
522531	4281.523	304.9795	2.56416E-20				
488972	14.03872						
071429							
Error	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*	
886243	95.54084	7.85E-49	95.94063549	100.0874224	95.94063549	100.0874224	
411913	-17.4637	2.56E-20	-1.839591353	-1.457964165	-1.839591353	-1.457964165	

The slide shows the Lower 95% and the Upper 95% columns in the Other statistics table selected.

Transcript

That's the quickest way that you can do that. One other thing that you will notice in Excel is that it will give you the upper bound and the lower bound of the confidence interval and if you don't mention that you 95%, the default is 95% and it will give it to you twice. So, if you say 99%, it's going to show you lower 99%, upper 99%, but also it will show you lower and upper 95%. So, I consider that to be a bug. So, I usually just go ahead and delete these because it's redundant.

Downtime Excel Worksheet (8 of 11) - Slide 85



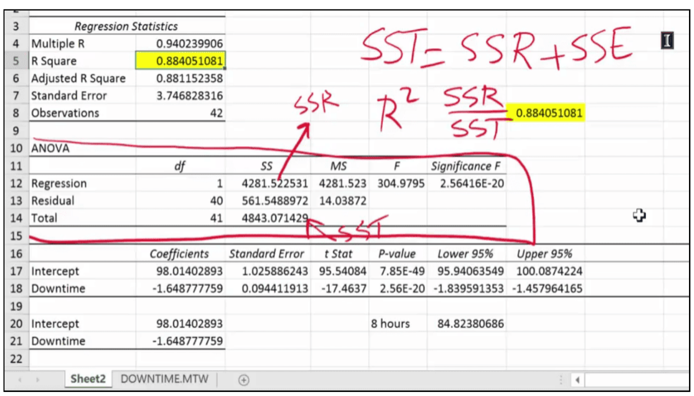
	A	B	C	D	E	F	G	H
1	SUMMARY OUTPUT							
2								
3	Regression Statistics							
4	Multiple R	0.940239906						
5	R Square	0.884051081						
6	Adjusted R Square	0.881152358						
7	Standard Error	3.746828316						
8	Observations	42						
9								
10	ANOVA							
11		df	SS	MS	F	Significance F		
12	Regression	1	4281.522531	4281.523	304.9795	2.56416E-20		
13	Residual	40	561.5488972	14.03872				
14	Total	41	4843.071429					
15								
16		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	
17	Intercept	98.01402893	1.025886243	95.54084	7.85E-49	95.94063549	100.0874224	
18	Downtime	-1.648777759	0.094411913	-17.4637	2.56E-20	-1.839591353	-1.457964165	
19								

The slide shows the R square value (0.8840) in the Regression statistics table, and the Downtime p-value (2.56E-20) in the Other statistics table highlighted. In addition, the following formula is presented: $\hat{y} = 98.01 - 1.648x$

Transcript

So, what you would see is that there is an intercept and there is a downtime. So, this is really my b0 and my b1. So, based on this, my regression equation is 98.01 − 1.648x, x being the downtime, how many hours the line has been down.

Downtime Excel Worksheet (9 of 11) - Slide 86



	A	B	C	D	E	F	G
3	Regression Statistics						
4	Multiple R	0.940239906					
5	R Square	0.884051081					
6	Adjusted R Square	0.881152358					
7	Standard Error	3.746828316					
8	Observations	42					
9							
10	ANOVA						
11		df	SS	MS	F	Significance F	
12	Regression	1	4281.522531	4281.523	304.9795	2.56416E-20	
13	Residual	40	561.5488972	14.03872			
14	Total	41	4843.071429				
15							
16		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
17	Intercept	98.01402893	1.025886243	95.54084	7.85E-49	95.94063549	100.0874224
18	Downtime	-1.648777759	0.094411913	-17.4637	2.56E-20	-1.839591353	-1.457964165
19							
20	Intercept	98.01402893			8 hours	84.82380686	
21	Downtime	-1.648777759					
22							

The slide shows the ANOVA table highlighted. Here, the Regression SS value (4281.522) is the SSR and the Total SS value (4843.071) is the SST. In addition, the following formulas are presented:
$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ and $R^2 = \frac{SSR}{SST}$. For this last formula, the R square value (0.8840) is placed as the outcome of R^2.

Below the Other statistics table is a small table that repeats the Intercept and Downtime coefficient information. A few cells to the right of the Intercept value (98.014) is a cell that says 8 hours - 84.8238.

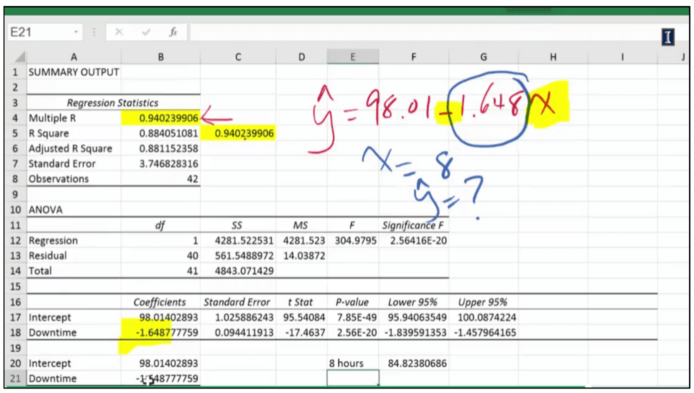
Transcript

So, one of the things that you want to look at is p-value for downtime is extremely small. One of the things that regression does is that the non-hypothesis is that x and y have no relationship and if you reject that, that means that you have found a variable that has a relationship with the other variable and when the p is so small we will reject the non-hypothesis, the non-hypothesis being they are not related at all. Then one of the things that we want to look at is R-squared. And R-squared is the percent of the variation that you can explain in the productivity because of downtime. So, 88.4% of productivity differences that we have collected in our data seems to be because of downtime.

So, let me explain R-squared a little bit in more detail. For that I'm going to go to this table, ANOVA table. What we see here is this is the sum of squares and this is the sum of squares due to regression variable and this is the sum of square due to total. The sum of square of the total shows the total variations that we see in our data. Some of this is coming from the regression variable and some of it is coming from the errors. Things that we have not counted for, we don't know what they are at this point. R-squared – R-squared is the ratio of SSR to SST, and the higher this value is, the more explanation we have found for the variations that we see, which means that we have a good predictive model. So, if I take that ratio right here, this is SSR divided by SST, and what you see here is exactly what you see here. That's what R-squared represents, the percentage of explained variations due to the regression variable.

Now, multiple R is your correlation. That is your correlation, and correlation is just simply square root of R-squared. If I take the square root of this value, you would get R. As you can see, these values are exactly the same, so R is the square root of R-squared. Correlation is a value that goes from zero to one or zero to negative one. The closer you are to one, the higher is the degree of correlation between the two variables. And the closer you are to zero, there is less correlation. The sign is simply telling you if they are directly related, directly related means that if one goes up, the other one also goes up, or negatively correlated or inversely correlated, which is if one goes up the other one goes down.

Downtime Excel Worksheet (10 of 11) - Slide 87



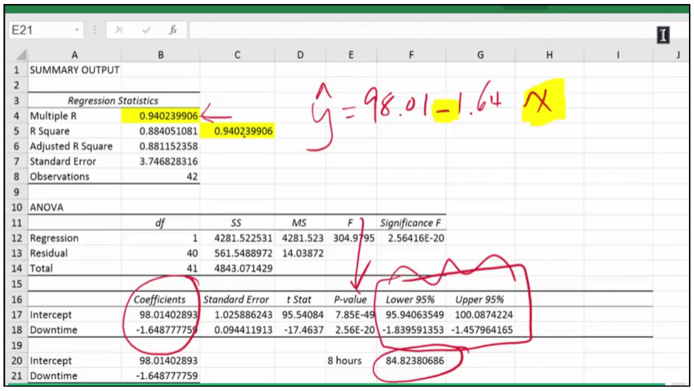
The slide shows the formula: $\hat{y} = 98.01 - 1.648x$, where $x = 8$.

Transcript

So, look at our downtime, as the number of hours of downtime goes up, what is happening to our productivity, it goes down. So, this is a negative correlation. A positive correlation would have had your points going this way. So, this is a negative correlation. How would you have known it by looking at this output? You would know this by looking at the sign, so the sign of the coefficient which you also see here is telling you if this is a positive correlation or a negative correlation. So, what does this mean now? That means that for every hour that you are down, our productivity goes down by another 1.6%, that's basically what it means.

So, if I told you that x is eight and I asked you to predict what would be the productivity, you simply would come and use this equation to answer that. And you can answer that, and I like just taking these values right here and copying it elsewhere and now I'm going to say for x of eight hours of downtime, what is my prediction? It's simply this value plus this value multiplied by eight, and that would be 84.8. So, if you're down in a given month for eight hours total, you expect your average productivity in that month to be 84.82. Now, let me make a point here, the coefficients that you see here are middle of a confidence interval so what you can see here is that you're also doing your confidence interval here.

Downtime Excel Worksheet (11 of 11) - Slide 88



The slide shows the coefficient column and the Lower 95% and Upper 95% in the Other statistics table highlighted.

[Download the Downtime Excel file (Refer to Regression Output - Third worksheet)]

Transcript

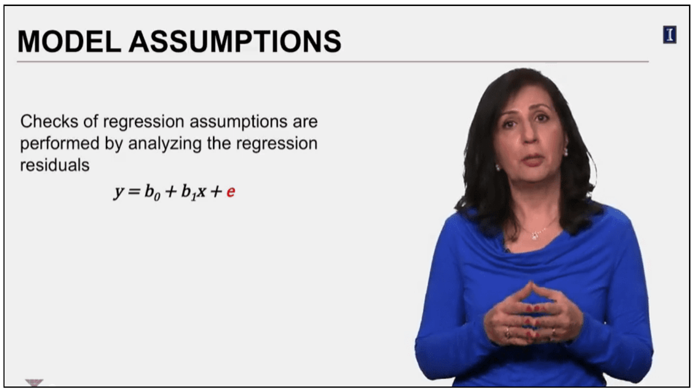
So, regression is bringing everything we have learned so far together. They bring hypothesis testing when we do p-value to check whether or not there is a significant relationship between variables that you have identified, it's going to give you a confidence interval for every variable so when I come up with this value, I have used the middle values, but this is just one estimation. We actually have to come up with a confidence interval for our prediction as well. The reason I am not showing that to you is because Excel really does not do a very good job of coming up with a predicted variable, we need to have another set of equations so I'm ignoring it, so what I'm showing you is a point estimation. This is a point estimation of if you had a month for eight hours of downtime, what would be the point estimate of its predicted productivity?

Lesson 3-5 Model Assumptions and Limitations

Lesson 3-5.1 Model Assumptions and Limitations

[Media Player for Video]

Model Assumptions - Slide 89



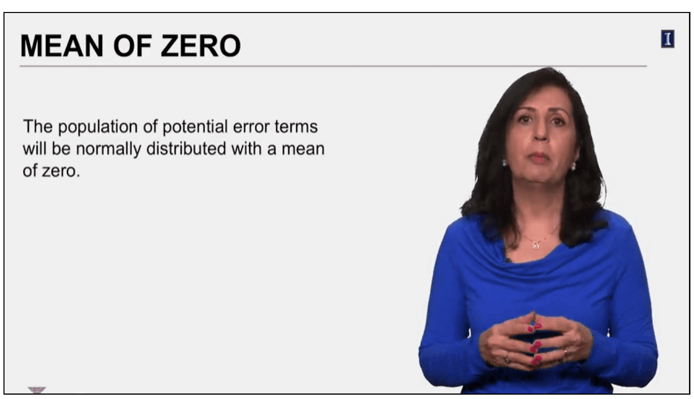
Checks of regression assumptions are performed by analyzing the regression residuals

$y = b_0 + b_1 x + e$

Transcript

Now that you have an understanding what simple linear regression analysis is, I would like to tell you about the assumptions needed so that the model can be applied correctly. One commonality about all assumption is that they all check for the models' validity by checking the error trends, also known as residuals, in the predictions. Recall that in the predictive value for y, for a given x, we will have some error. So, the actual y value will be the prediction plus some error value. Residuals are checked to make sure that simple linear regression is a valid model to use.

Mean of Zero - Slide 90

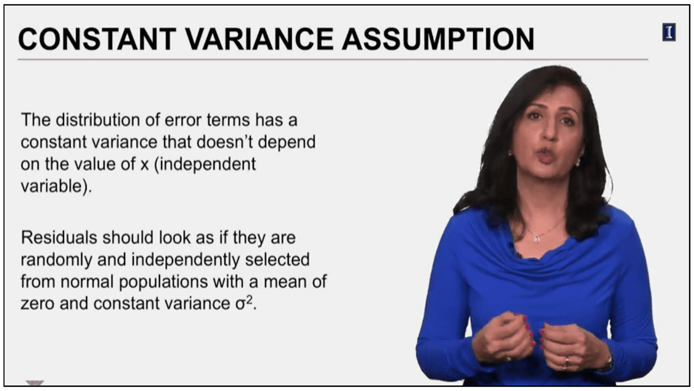


The population of potential error terms will be normally distributed with a mean of zero.

Transcript

If regression assumptions are valid, the population of potential error trends will be normally distributed with the mean equal to 0. This means that we will be over-predicting and under-predicting as a whole by equal amounts, so that when we add up all the error term, they will all cancel each other out and the mean error, will therefore, be 0, in showing that our model is not biased, to over-predicting or under-predicting.

Constant Variance Assumption - Slide 91



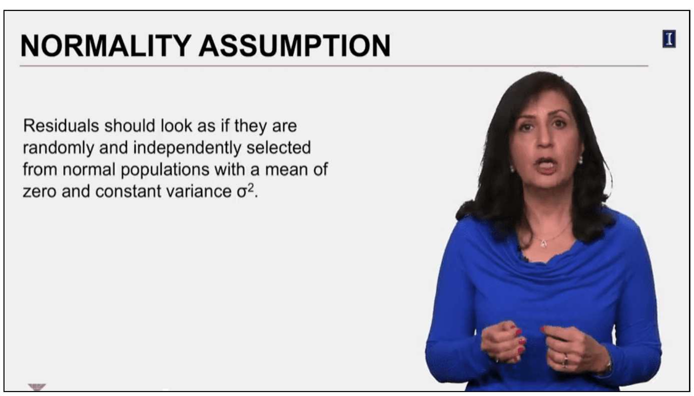
The distribution of error terms has a constant variance that doesn't depend on the value of x (independent variable).

Residuals should look as if they are randomly and independently selected from normal populations with a mean of zero and constant variance σ^2

Transcript

Second assumption is that assumption of constant variance. That means at any given value of x, the population of potential error term values has a variance that it doesn't depend on the value of x, the independent variable. As a result, the variance is constant across all values of x. Third, is the normality assumption, which builds on the first two assumptions. Residuals should look like they have been randomly and independently selected from normally distributed population have mean of 0 and a constant variance sigma square.

Normality Assumption - Slide 92

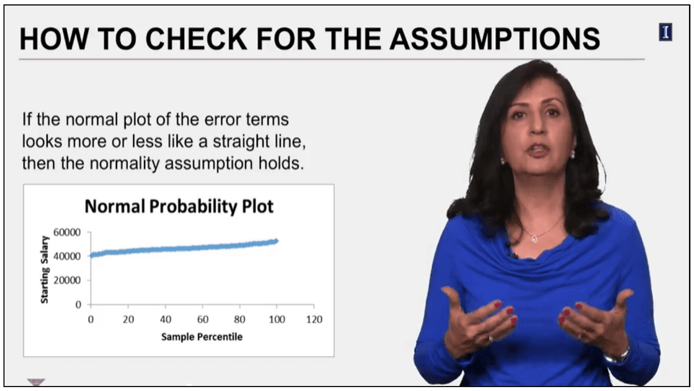


Residuals should look as if they are randomly and independently selected from normal populations with a mean of zero and constant variance σ^2.

Transcript

However, with any real data, the assumptions will not hold exactly. Mild departures do not affect our ability to make statistical differences in checking the assumptions. Their looking for pronounced departures from the assumption. As a result, we only require that the residuals approximately fit these descriptions.

How to Check for the Assumptions - Slide 93



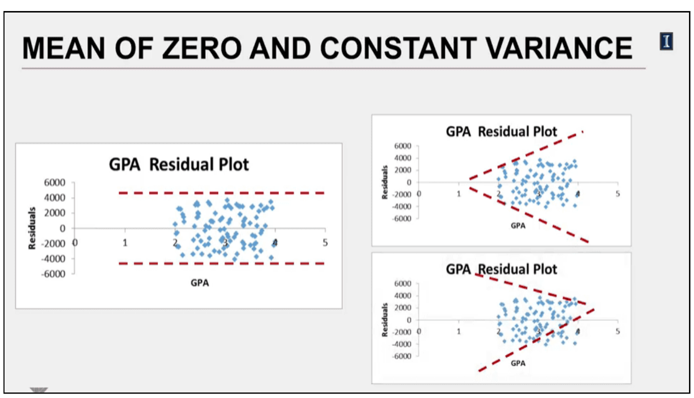
If the normal plot of the error terms looks more or less like a straight line, then the normality assumption holds.

The slide shows a graph that illustrates the normal probability plot. The x-axis is the sample percentile from 0 to 120 and the y-axis is the starting salary from 0 to 60000. The scatter plot is almost a straight line with intercept 40000 in the y-axis and a very small positive slope.

Transcript

The assumptions are checked to plotting of the error terms, if the normal plot of the error terms look, more or less, like a straight line, then normality assumption holds. Here's a normal probability plot for error terms in the effective GPA on starting salary study I showed you in the last lesson. The error terms line up nicely and look like a straight line, so the normality assumption holds in this example. For the same example, here's the distribution of error terms around the center line. This represents a mean of 0.

Mean of Zero and Constant Variance - Slide 94

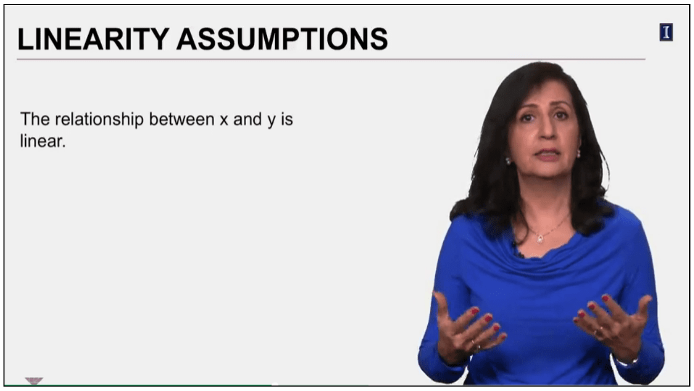


There are three residual plots with the x-axis as GPA from 0 to 5 and the y-axis as residuals from −6000 to 6000. The left plot is equally distributed among the x-axis. The data points above 0 and below 0 are almost equal. The trend line of data points above 0 and below 0 are two parallel lines with the x-axis. In the right top plot, the residual data points of GPA start on 2 and have a narrower distribution than those at GPA higher than 2. The trend of residuals above 0 is an increase line, while for residuals below 0 the trend is a decrease line. The bottom right plot is the reverse of the top right one. The residuals at GPA around 2 is sparser distributed than those at higher GPA. Thus, the trend line for above 0 residuals is decrease line, while that for below 0 residuals, the trend is an increase line.

Transcript

At first you can see that we see about the same number of error terms above and below the 0 line, which will give us an overall error of 0. So, mean of 0 assumption holds as well. Now look at the shape of the distribution of these errors. We see that the residuals bearing up and down within a contained horizontal band. The means the assumption of constant variance. The constant variance would have been violated if the plot of errors fanned out, starting out small and getting larger, which would have meant an increasing or the opposite occurs, starting out large and decreasing.

Linearity Assumptions - Slide 95

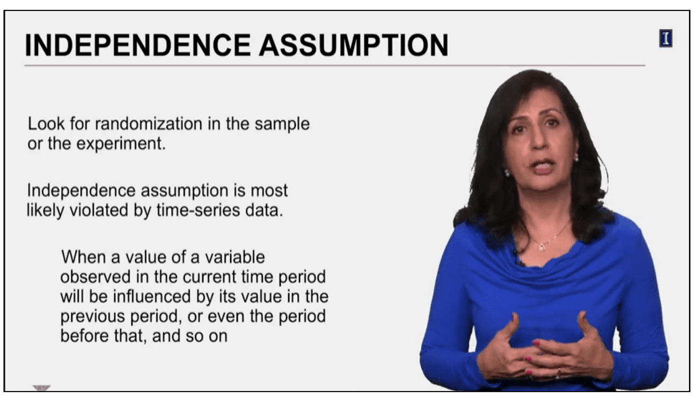


The relationship between x and y is linear.

Transcript

So again, with visual inspection of these plots, we can check for the constant variance assumption. And finally, is the linearity assumption, which is the condition that is satisfied if the scatter plot of x and y looks straight.

Independence Assumption - Slide 96



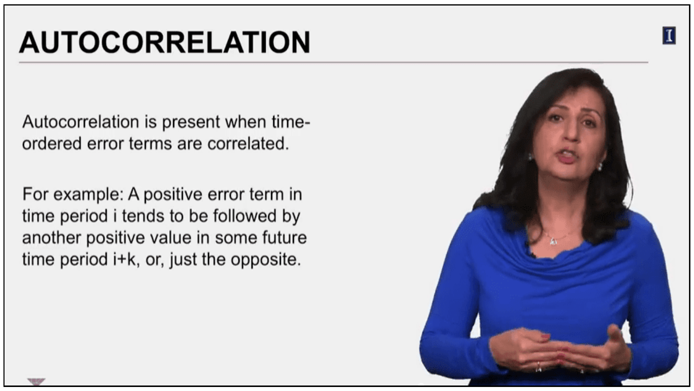
Look for randomization in the sample or the experiment.

Independence assumption is most likely violated by time-series data. When a value of a variable observed in the current time period will be influenced by its value in the previous period, or even the period before that, and so on.

Transcript

Then, it's the independence assumption. That is any one value of error term is statistically independent of any other value of the error term. This really is an important assumption. An independent variable must be truly independent. The independence assumption is usually, only violated when the data or time-series data. That is because assumption is positional, its value depends on order of the data. When the data is not time-series, it has no meaningful order, so, any order is acceptable. Independence is violated, then a valuable variable observed in the current time period will be influenced by its value in the previous period, or even the period before that, and so on. This is also known as autocorrelation.

Autocorrelation - Slide 97



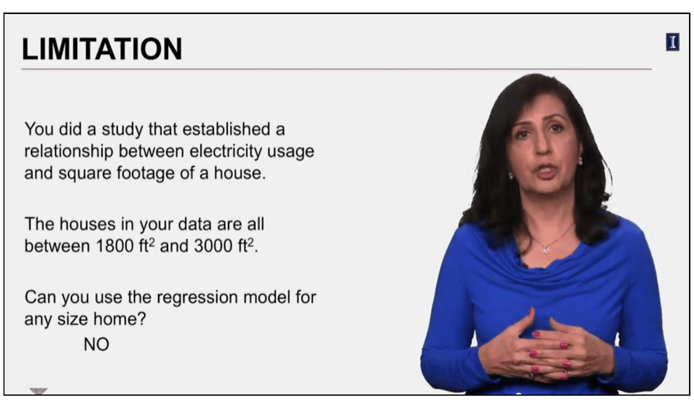
Autocorrelation is present when time-ordered error terms are correlated.

For example: A positive error term in time period i tends to be followed by another positive value in some future time period i+k, or, just the opposite.

Transcript

Positive autocorrelation, which is more common, is when a positive error term in time period i, tends to be followed by another positive value in some future time, i + k. For example, if you're looking at money spent on leisure and travel, you know that we tend to do more of this in the summer months. So, the fact that the spending starts to go up in June, will also mean July will go up, and so on. So, there's some time series impact and positive autocorrelation between months of summer. And they expect these patterns of spending to reappear again in 12 months. Of course, we can also have negative autocorrelation which is just the opposite. In negative error term in time-period i, tends to be followed by a negative value in some future time, i + k. Linear regression is not appropriate for these types of data.

Limitation - Slide 98



You did a study that established a relationship between electricity usage and square footage of a house.

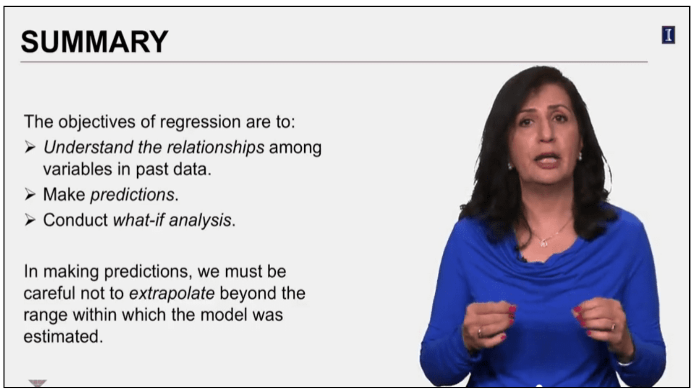
The houses in your data are all between 1800 ft^2 and 3000 ft^2.

Can you use the regression model for any size home? → NO

Transcript

If the data, more or less, doesn't violate assumptions mentioned, then the linear regression can be used. Then you should be mindful of how to apply the model. The model is only valid for the range of data you have analyzed. Consider this case. We did a study which established a relationship between electricity usage and houses square feet. You have data collected for the house size, in square feet, and how much kilowatt hours of electricity is used per month. The houses in your data are all between 1,800 square feet and 3,000 square feet. The power company sees a new housing development coming up and wants to make sure it will have enough capacity for the additional demand for electricity needed for this new development. Homes in this development will be between 5,000 to 7,500 square feet. Can the power company use the model we developed? The answer is no. We must be very careful not to extrapolate beyond the range that was in our data when we developed the regression equation.

Summary (1 of 3) - Slide 99



The objectives of regression are to:

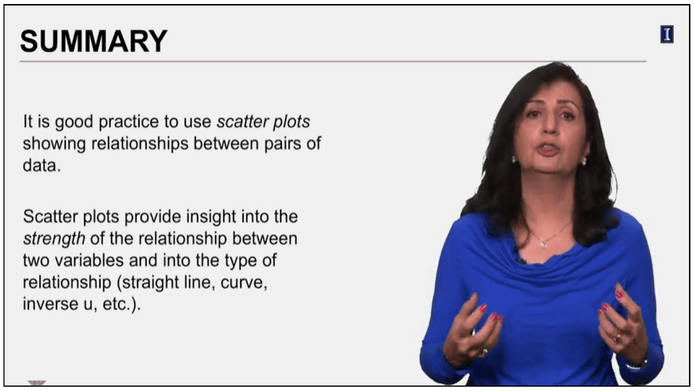
- *Understand the relationships* among variables in past data
- Make *predictions*.
- Conduct *what-if analysis*.

In making predictions, we must be careful not to *extrapolate* beyond the range within which the model was estimated.

Transcript

So, in summary, let me remind you that the objectives of regression are to understand the relationship between variables and past data to make predictions and conduct what-if analysis. And making predictions, we must be careful not to extrapolate beyond the range in within which the model was estimated.

Summary (2 of 3) - Slide 100



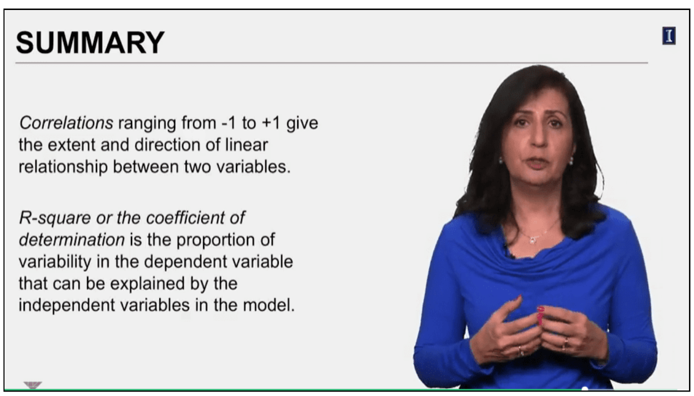
It is good practice to use *scatter plots* showing relationships between pairs of data.

Scatter plots provide insight into the *strength* of the relationship between two variables and into the type of relationship (straight line, curve, inverse u, etc.).

Transcript

Prior to estimating regression model, it is a good practice to use scatter plots showing relationship between pairs of data. Scatter plots provide insight into the strength of the relationship between two variables and into the type of relationship, straight line, curve, inverse, so on and so on.

Summary (3 of 3) - Slide 101



Correlations ranging from −1 to +1 give the extent and direction of linear relationship between two variables.

R-square or the coefficient of determination is the proportion of variability in the dependent variable that can be explained by the independent variables in the model.

Transcript

Correlation ranging from negative 1 to positive 1 give the extent of linear relationship and the direction of linear relationship between the two variables. R-squared, the coefficient determination, is the proportional variability in the dependent variable that can be explained by the independent variables.

Word of Caution - Slide 102

Correlation is not the same as causation.

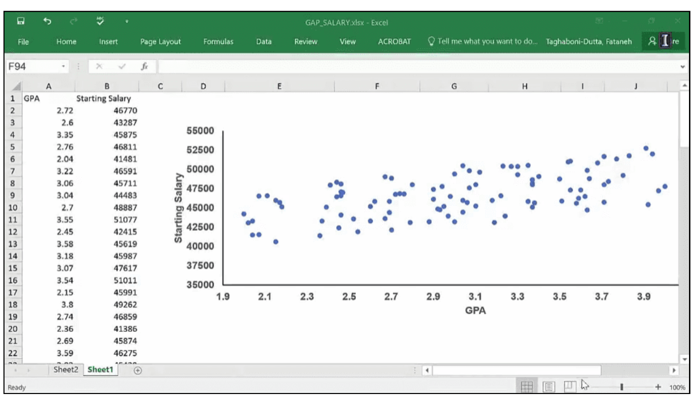
Transcript

R-squared values are bound by 0 and 1. While it's very exciting to find and describe the relationship between two variables allows us to make predictions, we should not confuse correlation with causation. Proving causation requires evidence far greater than most of these can do or need to do. Just remember, how long it took to establish smoking as a cause for lung cancer. We knew that smoking and cancer correlated for a long time, but to establish it as a cause took far greater time. You can see the same argument today with global warming. While most everyone agrees that the climate is changing, the argument and its cause is not as well accepted. So, when you do regression, don't claim that you have found the cause. Just knowing the correlation on its own gives, us a great ability to predict. This allows us to change outcomes when we don't like what will be happening by changing the values of the independent variable.

Lesson 3-5.2 Residual Analysis in Excel

[Media Player for Video](#)

GPA Salary Excel Worksheet (1 of 8) - Slide 103



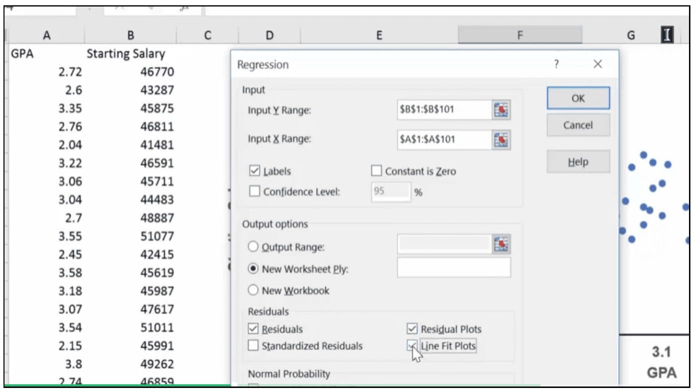
The slide shows the GPA Salary data set. The data includes two columns: GPA and Starting salary. In addition, there is a scatter plot with the x-axis as the GPA from 1.9 to 4.0 and the y-axis as the Starting salary from 35000 to 55000. The data points form a positive correlation, where as the GPA increases, the starting salary gently increase.

[Download the GPA Salary Excel file (Refer to Data Scatter Plot - First worksheet)](#)

Transcript

In this video, I'm going to show you how to do the residual analysis so we can check the underlying assumption in a simple linear regression. This is the example that I have already shared with you in my lectures. This is the impact of GPA on starting salary of the students. And we have plotted the relationship between the two, and it pretty much looks like a straight line. It's not as sharp of an increase as one would like to see maybe, but it is still an increasing relationship between GPA and starting salary. We can get that also from the data analysis. So, let me just do that. Let me run the regression by going to data analysis, pick regression, and when the regression comes up, you have to first give it the y-range.

GPA Salary Excel Worksheet (2 of 8) - Slide 104



The slide shows what to include in the Regression pop-up window. Once going to the Data tab in the main menu, selecting Data Analysis, and choosing the Regression option, another pop-up window appears. The window consists of four sections: Input, Output, Residuals, and Normal probability. In the Input section, the Input y range is $B$1:$B$101, the Input x range is $A$1:$A$101, the Labels checkbox option is selected, and the Confidence level is set to be 95%. In the Output section, the New worksheet ply radio option is selected. In the Residuals section, the Residuals checkbox, the Residual Plots checkbox, and the Line Fit Plots checkbox are selected.

Transcript

So, y is what you're trying to predict. And, here is the starting salary. So, I'm going to put my cursor on the first cell, control shift down and pick the whole thing. Now, remember, if you mix up x and y, there is nothing that Excel will do to alert you to that. It will give you an answer, it will just be a wrong answer. So, be very careful about how you're going to put your x's and y's because it's not going to look odd to you at all. So, the next thing I want to do is put my GPA, and that's in the column A. I have labels, so I'm going to click on that. I'm going to put it in a new worksheet and I'm going to now ask it to plot residuals, residual plots, line fit plots, and normality plots, and click okay. And here's what we get.

GPA Salary Excel Worksheet (3 of 8) - Slide 105

Regression Statistics	
Multiple R	0.580085
R Square	0.336499
Adjusted R	0.329729
Standard E	2236.609
Observatio	100

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	1	2.49E+08	2.49E+08	49.70136	2.53E-10
Residual	98	4.9E+08	5002421		
Total	99	7.39E+08			

	Coefficients	*andard Errc*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Uppe*
Intercept	37916.17	1258.16	30.1362	2.32E-51	35419.39	40412.95	35419.39	40
GPA	2904.709	412.0202	7.049919	2.53E-10	2087.068	3722.35	2087.068	3

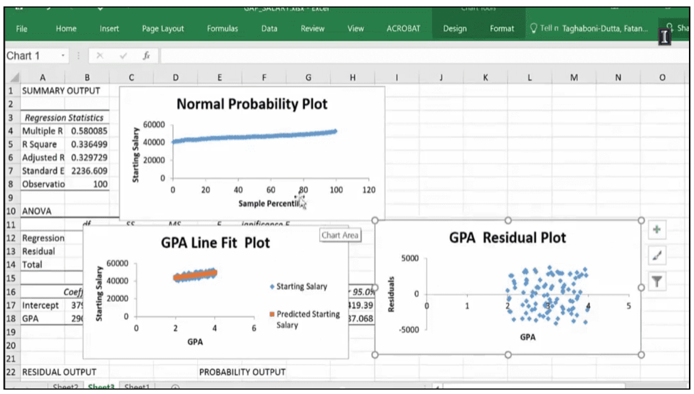
The slide shows the summary output distributed into three tables with their respective values. The first table is the regression statistics, the second table is the ANOVA, and the third table are other statistics that compare the Intercept with the GPA.

[Download the GPA Salary excel file (Refer to Output - Second worksheet)]

Transcript

First of all, you will see that the R-squared is only 33.65. So, 33.65% of the salary variations is explained by the GPA. Also, you would see that for every point your GPA goes up, your salary goes up by 2,904. Remember, that means if your GPA goes up from 2.0 to 3.0, you're going to get a 2,900 boost in your salary, so that's why the rise wasn't so much because we have a lot of numbers between two and three.

GPA Salary Excel Worksheet (4 of 8) - Slide 106

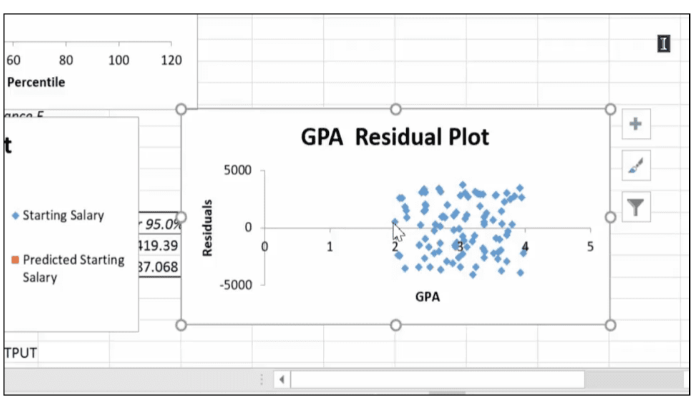


The slide shows three residuals plots: normal probability plot, GPA line fit plot, and GPA residual plot. For the normal probability plot, the x-axis is the sample percentile from 0 to 120 and y-axis is the starting salary from 0 to 60000. The scatter plot is almost a straight line with intercept 40000 in the y-axis and a very gentle positive slope. For the GPA line fit plot, the x-axis is GPA from 0 to 6 and y-axis is starting salary from 0 to 60000. All the original data points are marked blue and the predicted regression line is marked red. The red regression line almost covers all the blue data points. For the GPA residual plot, the x-axis is GPA from 0 to 5 and y-axis is residuals from −5000 to 5000. The residuals are equally distributed among the x-axis. The data points above 0 are almost equal as those below 0.

Transcript

So, these are the plots that will show us whether or not we have a linear regression that fits the assumptions or not. So, when you have a normality plot, then you see these points and it looks pretty much like a straight line. The normal distribution is not being violated, so this is good. The other thing is that residual plot shows the variability of the errors.

GPA Salary Excel Worksheet (5 of 8) - Slide 107

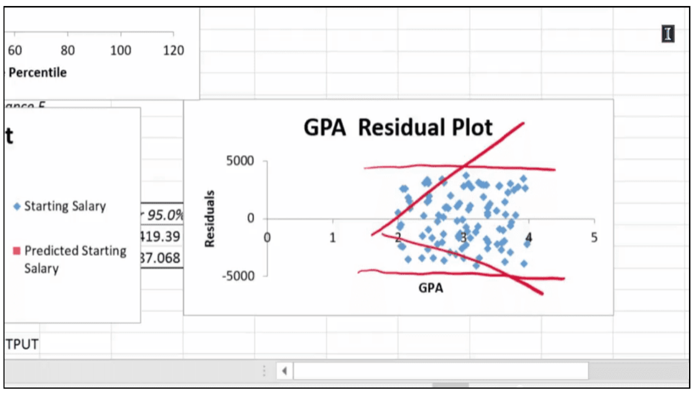


The slide shows the same GPA residual plot as [Slide 106 GPA Salary Excel Worksheet (4 of 8).](#)

Transcript

Remember, sometimes you're going to overpredict and sometimes you're going to under predict. So, these are all the things that have been overpredicted and these are all the things that have been underpredicted. And if you just visually look at this, then these values and these values seems to be about the same which means, if I sum them up, I should get that zero. So, one of the things that we need is that the errors as a whole will cancel each other out to a mean of zero. And, visually inspecting that, you get the sense that that will be the case. It wouldn't have been the case if the points were a lot of them were on the top or vice versa, a lot of them were at the bottom. But, here, they seem to be evenly split up. The other thing is that the variabilities themselves seem to be kind of bounded by a straight line. So, if I can think of a straight line, if I can think of a straight line, it seems to be they're bounded, these are the boundaries by which they're bouncing around. And so, therefore, it's telling me that it has a constant variance.

GPA Salary Excel Worksheet (6 of 8) - Slide 108

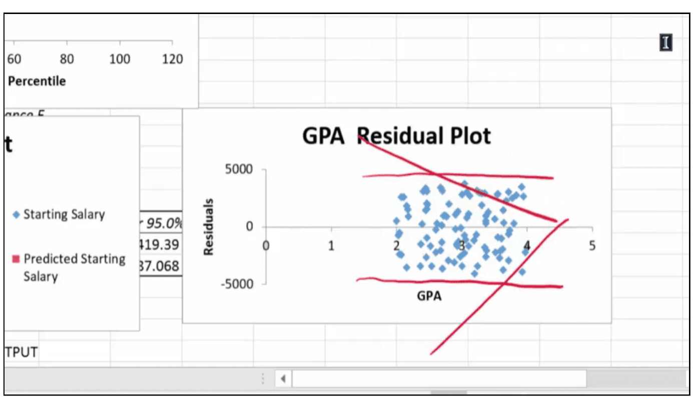


The slide shows the GPA residual plot with two parallel lines on the plot. Assuming that the trend of residuals above 0 is increasing and the trend of residuals below 0 is decreasing, two additional lines are drawn: an increasing red line on the residuals above 0 and a decreasing red line at residuals below 0.

Transcript

If this plot looked like this, that was fanning out. Or looked like this, then we would have said that the constant variance is violated.

GPA Salary Excel Worksheet (7 of 8) - Slide 109

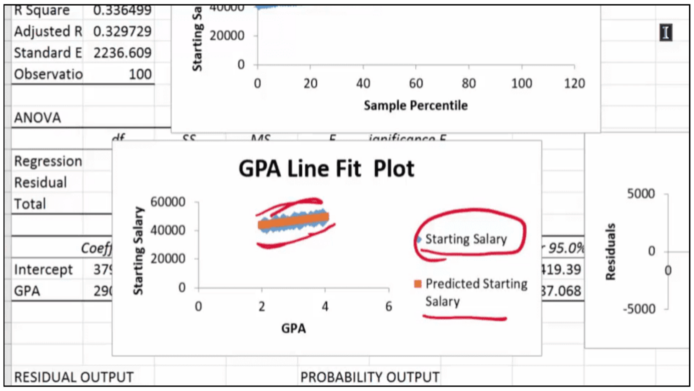


The slide shows the GPA residual plot with two parallel lines on the plot. Assuming that the trend of residuals above 0 is decreasing, and the trend of residuals below 0 is increasing, two additional lines are drawn: an increasing red line on the residuals below 0 and a decreasing red line at residuals above 0.

Transcript

And this is the line fit.

GPA Salary Excel Worksheet (8 of 8) - Slide 110



The slide shows the same GPA line fit plot as Slide 106 GPA Salary Excel Worksheet (4 of 8). Here, there is one line above and one line below the line fit. These lines show that the points are evenly divided above and below the line fit.

Download the GPA Salary Excel file (Refer to Plots - Third worksheet)

Transcript

Again, what you will see is that the orange line here is representing the predicted salaries, the predicted salaries is the orange line. And the starting salary is in blue. So, if you, again, look at this, you can see that we have, we seem to be about the same. We have lots of points underneath, but we also have some points on the top. And they seem to be evenly divided. So, this is what you want. If your data fits, you should have randomness in terms of your errors. It shouldn't be always overpredicting or, majority of time, overpredicting or underpredicting. Or you see any kind of a pattern like seasonality pattern to it. So, these are things that we look at. We look at the residual plots to see if any of the underlying assumptions that are necessary for simple linear regression is being violated or not. If the – it's being violated, then you really should not be using the model.